

Calibration-Aware Reinforcement Learning for Large Language Models: A Survey of Objectives, Optimization, and Decision-Making

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Abstract

Large language models increasingly produce confidence reports, predictive distributions, and structured decisions that determine whether a system answers, abstains, retrieves evidence, or spends additional computation. Reinforcement learning can improve these signals, but it can also change the answers being assessed, the meaning of the training target, and the effective scoring incentive. We survey calibration-aware reinforcement learning through the connection between probability targets, objectives, implemented updates, and decision use. The synthesis combines a comparison of representative methods, with claims traced to primary sources, and three mechanisms for interpreting divergent results. Classical proper-score geometry separates mean capability, the distribution of conditional success probabilities, and reporting regret in joint answer–confidence training. Group transformations, parameter-dependent rewards, and finite-ensemble scoring distinguish an ideal population objective from the update actually implemented. Cost-sensitive decisions distinguish useful ordering from probability magnitudes that can be reused across operating points. These distinctions organize comparisons with post-hoc calibration, supervised reporting, selective ranking, direct action learning, and numerical distribution prediction. We provide a method landscape, an evaluation contract, and a research agenda for separating reporting improvements from policy changes, estimator effects, and threshold adaptation. The resulting perspective treats calibration as a probability claim whose conditions must remain explicit throughout learning and deployment, and identifies when RL supplies an intervention that simpler calibration methods do not.

1. Introduction and scope

A language model’s confidence becomes consequential when another component acts on it. Reported correctness probability 0.9 may trigger automatic acceptance, whereas 0.5 may trigger retrieval; a numerical forecast may determine resource allocation. Typed-decision services such as TypeSafe’s Jev expose decisions and probability fields directly to downstream software [1]. Their numerical meaning must match the consuming decision, beyond fluent justification or valid formatting. Reinforcement learning (RL) can change both the report and the behavior about which it is made.

The literature supplies apparently conflicting observations. The GPT-4 technical report shows worse answer-option calibration after post-training on the same MMLU subset [2]. A study of reasoning models finds improved discretized verbal confidence, alongside task- and interface-dependent exceptions [3]. Another comparison reports lower average expected calibration error (ECE) after GRPO or DPO than at initialization, while a supervised baseline performs better than either [4]. These results concern different probability readouts, supervision, and evaluation populations. Together, they motivate separating the training intervention from the probability interface and evaluation population. Across the checkpoints studied, distillation and RL generally improve verbalized calibration on reasoning tasks while reducing “I don’t know” responses on factuality benchmarks [5]. The Hallucination Tax study addresses a related behavioral outcome: reinforcement fine-tuning can reduce refusal on unanswerable questions, even though that study

evaluates answer/refusal behavior rather than numerical probability calibration [6]. Holding an answer fixed while changing its chat-role framing can also change confidence [7].

Three more precise divergences motivate this survey. **Reporting and capability can diverge:** an objective may reward honest confidence while preferring a different distribution of success across questions. **The stated score and implemented learning can diverge:** normalization, token masks, finite rollouts, and direct reward derivatives determine whether training follows the intended objective. **Ordering and decision value can diverge:** a score can improve selective ranking while losing the numerical interpretation required by a fixed probability threshold. Each divergence calls for a different comparison and potentially a different remedy. Collecting their metrics under a single “better confidence” label obscures those choices.

Our thesis is that calibration-aware RL requires preserving the meaning of a probability claim across target construction, objective design, implemented learning, and decision use. Method-level comparisons must distinguish fixed-event reporting, joint answer–confidence objectives, and action rewards. Even a full probability-vector output may be trained by scoring only its selected component [8].

1.1. Contribution and relation to existing surveys

Existing surveys cover confidence estimation and calibration [9], broad RL objectives and optimization [10], uncertainty as an active signal in reasoning and agents [11], uncertainty quantification and its applications [12], and abstention [13]. These surveys establish the broader estimation and control landscape.

Two related manuscripts with overlapping authorship warrant explicit comparison. *Confidence as Control* [14] organizes confidence-related signals by source, decision unit, and control role across training and deployment, including preference and policy optimization. The unpublished companion manuscript *Beyond Natural-Language Interfaces: Decisions, Scores, and Predictive Distributions* examines how predictive reports are produced, transformed, and delivered to consumers, with attention to supervision, access, and computation. These works share probability semantics, calibration–ranking distinctions, selected literature, and consumer-sensitive evaluation. The present survey concentrates on the connection between probability-learning incentives, implemented updates, and decision losses: joint answer–confidence objectives, group transformations, direct reward derivatives, finite-ensemble scoring, and the reuse of reports across operating points. The distinction concerns the organizing question and detailed synthesis; overlap extends to statistical foundations and training comparisons, as well as the decision interfaces in Sections 5–6. Appendix E documents these relationships.

Figure 1 and Table 2 map the methods by probability object, interface, feedback, objective, update, and decision, with protocol details in Appendix B. **Section 3 develops the reporting–capability decomposition**, using an exact construction to separate truthful reporting from mean-accuracy preference. **Section 4 compares group transformations, direct reward derivatives, and finite-ensemble scoring**, with a comparator guide in Table 3. **Section 6 supplies an evaluation contract and within-study comparisons** in Tables 4–5; Section 7 identifies four open questions and the evidence needed to distinguish explanations. The identities draw on established statistics [15], [16]; the contribution is their integration with concrete learning protocols and decision requirements.

The synthesis retains favorable trade-offs: lower average accuracy can accompany a larger safely answerable subset, ranking can suffice for fixed coverage, and direct action learning can suit a single cost regime. Probability estimation earns its additional complexity when numerical meaning is needed across decisions, information states, or operating points.

1.2. Scope and method

We review calibration-aware RL together with supervised, preference-based, distillation, and post-hoc comparators, supported by foundational results, post-training diagnostics, and relevant decision interfaces. Selection combines targeted primary-source searches and citation tracing. We verify source identity and check claim-bearing methods, equations, experimental tables, and implementation details; recorded versions and reading ranges accompany the source collection. The literature snapshot is 20 September 2026.

The collection contains 98 academic records and eight primary web sources; Table 2 and Appendix B organize the learning and comparator protocols within it. Reported model results retain their original study populations and protocols; analytic constructions and public-code observations are identified separately.

Calibration-aware reinforcement learning

A field map of objectives, learning mechanisms, and decision use



[†] Non-RL comparator or classical theory. [‡] Industrial interface. Methods may recur across views.

Jev[‡] [1] §5.6 Typed serving is an interface case; the full RL training recipe is undisclosed.

Figure 1: Faceted taxonomy of calibration-aware RL and its comparators. Three complementary illustrated facets organize objectives, learning mechanisms, and decision uses; a method may recur across facets. The scenes are conceptual illustrations, not empirical results or a sequence of training stages. Method names and section labels are native, clickable text linked to bibliography entries and their discussion. Detailed distinctions appear in Sections 3-6 and Table 2. Jev is an industrial interface case; its inclusion does not establish a proprietary training recipe.

2. Targets and comparison framework

A calibration claim begins with an event and the information available when its probability is reported. An identical output, such as “0.8,” can describe the correctness of an answer, the success rate of a sampling policy, or agreement within an annotator population. These quantities require different supervision and support different decisions. We therefore compare methods by the claim their output makes, then ask how that claim connects to the objective, implemented update, and evaluation (Figure 2). The resulting organization links the established statistical objects to the learning choices compared throughout the survey.

2.1. Events, information, and reports

Let X denote an input, A a generated answer or action, and $Z \in \{0, 1\}$ a declared binary outcome. For answer confidence, Z may indicate whether A passes a specified correctness test. For forecasting, it may indicate whether an external event occurs. Write $\mathcal{I}$ for the declared statistical conditioning information and

$$p_{\mathcal{I}} = \mathbb{E}[Z \mid \mathcal{I}], \quad C \in [0, 1]$$

for the corresponding conditional probability and its report. Conditional scoring arguments require the report to be measurable with respect to the stated information. The protocol must separately record the evidence actually available to the model and must not supply feedback that arrives after the decision being evaluated.

For a deterministic correctness rule, conditioning on the complete prompt and answer makes the mathematical label probability zero or one; computational difficulty alone does not change this fact. A nondegenerate target therefore requires a stated coarser conditioning, such as a report or restricted representation, uncertainty over latent or stochastic labels, or a stochastic answering procedure. Population calibration given a score remains meaningful even for deterministic benchmark labels. A post-answer reporter can inspect the realized answer and reasoning; a pre-generation capability estimator cannot. Report timing and statistical conditioning must both accompany the claim.

For a fixed generation and decoding policy π , capability can instead be written

$$p_{\pi}(X) = \Pr_{\pi}(Z = 1 \mid X),$$

where the probability averages over the declared sampling procedure and any outcome randomness. Repeated rollouts estimate this policy-specific success rate. They do not directly label the correctness of a particular answer already selected. Capability estimates used for repeated-attempt budgets consequently depend on the decoder and assumptions about repeated samples [36]. Changing the policy changes this target even if the benchmark questions remain fixed.

2.2. The principal objects

Table 1 separates output form from statistical meaning. A vector of numbers does not establish full-distribution supervision, and a generated numeral is not necessarily an event probability.

Table 1: Probability objects, conditioning and representative settings.

Object	Quantity and relevant information	Representative setting
External binary event	$\Pr(Z = 1 \mid \mathcal{I})$, for an outcome mechanism held fixed during the reporting comparison	Stochastic-outcome scoring experiments [30]
Class-label posterior	Q_k intended to describe $\Pr(Y = k \mid \mathcal{I})$; distinguish the complete vector from its selected component	Decision-token calibration [31]; verbalized vectors in RLVP [22]

Object	Quantity and relevant information	Representative setting
Realized-answer correctness	$\Pr(A \text{ is correct} \mid \mathcal{I}, A)$, possibly including the answer’s reasoning trace	$P(\text{True})$ evaluation of a proposed answer [39]; joint answer/confidence learning [19]
Pre-generation capability	$\Pr_{\pi}(Z = 1 \mid X)$, under a named generation and decoding policy	$P(\text{IK})$ under a specified sampling policy [39]; inference-budget prediction [36]
Population preference	$\Pr(B \succ A \mid X, \text{annotator process})$, allowing persistent disagreement	Preference-distribution learning [38]
Numerical outcome statistic	A conditional mean or another declared functional of a numerical target	REAL’s mean-oriented objective [26]
Numerical outcome distribution	$F(t \mid \mathcal{I}) = \Pr(Y \leq t \mid \mathcal{I})$, beyond a point prediction	Distributional regression with empirical CRPS [27]
Action-selection probability	$\pi(a \mid s)$, describing how often the policy selects an action	Correct/abstain/wrong action rewards [28]

Kadavath et al. established the useful $P(\text{True})/P(\text{IK})$ distinction: evaluating a supplied answer and predicting success before an answer is supplied require different conditioning [39]. Their $P(\text{IK})$ training updates the language model and a value head from sampled correctness labels, placing supervised capability estimation alongside later RL approaches. Verbalized uncertainty training [40] and elicitation comparisons [41] add another axis: the same answer can be scored through label logits, a numerical report, or a prompted self-evaluation. Section 4 compares the learning choices for these interfaces.

The final row is particularly important for decision-oriented LLMs. Increasing the probability of selecting a useful action does not, by itself, make that probability an estimate of the action’s correctness. Similarly, a preference distribution measures the specified population’s choices, rather than uncertainty about a uniquely correct winner. In the principal experiments of [38], that population is simulated through persona-conditioned annotators. The target remains meaningful, but its population must accompany the claim.

The same separation applies inside a reporting interface. ConfTuner defines a distribution q_i over candidate confidence numerals and directly minimizes expected squared error of their numerical values [32] (Eq. 1). The numeral distribution is a mechanism for producing a report; it is not itself the answer’s correctness posterior. Conversely, emitting a full class vector does not mean every component was scored. A selected-answer reward can support empirical improvements in multiclass metrics without inheriting strict propriety for an unrestricted vector [8] (Table 6).

2.3. Calibration is one property among several

For binary correctness, we use **score calibration** to mean

$$\mathbb{E}[Z \mid C] = C \quad \text{almost surely.}$$

Guo et al.’s confidence-calibration definition conditions on confidence alone [33] (Eq. 1). If $\hat{Y}$ is the predicted class, class-conditional top-label calibration additionally requires $\mathbb{E}[Z \mid C, \hat{Y}] = C$. Full-vector calibration requires $\mathbb{E}[\mathbf{1}\{Y = k\} \mid Q] = Q_k$ for each class. When the prediction and its confidence are determined from that vector, the full-vector condition implies the corresponding selected-class conditions. The reverse implications generally fail.

None of these equalities establishes correctness conditional on all available input information. A constant report equal to the population success rate is score-calibrated but cannot distinguish difficult from easy cases. Resolution concerns such informativeness; discrimination concerns ordering correct above incorrect answers. A strictly increasing transformation preserves scalar rankings, including AUROC under consistent tie handling, while generally changing calibration. Equal AUROC values alone, however, do not establish equal rankings or identical operating curves.

For binary correctness Z and a scalar report C , the population Brier decomposition makes this distinction precise. Write $m(C) = \mathbb{E}[Z \mid C]$ and $\mu = \mathbb{E}Z$. Under their fixed joint distribution [42],

$$\mathbb{E}(C - Z)^2 = \underbrace{\mu(1 - \mu)}_{\text{uncertainty}} - \underbrace{\text{Var}(m(C))}_{\text{resolution}} + \underbrace{\mathbb{E}(C - m(C))^2}_{\text{squared reliability error}}.$$

The identity conditions on the exact report; estimating its terms with bins introduces a separate approximation. Resolution measures information about correctness retained by the score. Under calibration it becomes $\text{Var}(C)$. A change in the answer-generating policy can also change the outcome base rate and hence the uncertainty term.

Sharpness describes the concentration of predictive distributions without reference to realized outcomes. Gneiting, Balabdaoui, and Raftery formulate maximizing sharpness subject to calibration as a guiding principle for probabilistic forecasting [43]. This distinguishes useful concentration from an unconditional preference for certainty. Proper scores evaluate broader predictive quality: their improvement can reflect reduced reporting error, more informative predictions, or a changed outcome population. Strict propriety identifies a unique truthful expected-score maximizer within a specified report class; it does not establish that a finite neural optimization attains it [15].

Coverage must also name its object. Selective-answering coverage is the fraction of queries answered, and selective risk measures error among the retained predictions [44]. Risk-coverage curves and their area summarize a selector across operating points; they combine the underlying prediction errors with their ordering [45]. Predictive-set coverage instead measures the probability that a set contains the target. Conformal methods establish a marginal coverage statement under specified sampling assumptions; conditional or subgroup coverage requires stronger provisions [46]. These primary formulations identify the property each training or selection procedure must evaluate.

A further use of the word calibration concerns a generative distribution over semantic facts. Kalai and Vempala’s hallucination lower bound assumes a sparse fact space and limited information about unseen arbitrary facts [47]. Its calibration object is the generated fact distribution, whereas the correctness reports in this survey are evaluated against their declared answer or event targets. The result therefore provides a distinct statistical perspective on hallucination; transferring its lower bound to reported correctness confidence requires a model connecting the two objects.



Figure 2: A probability claim has a context. The same reported number can refer to three different targets: an external event, a realized answer, or the success of a sampling policy. The lower workflow connects the target to feedback, an objective, an implemented update, and a decision. Each connecting question identifies a condition that must be checked rather than an automatic guarantee. Section 2 uses this reading aid to organize established distinctions [39], [11], [15].

2.4. How the literature is organized

Figure 1 uses three complementary views, rather than assigning each paper to one exclusive family. A method can share an objective with one neighbor, an update with another, and a deployment use with a third. Table 2 synthesizes the families along common statistical axes: what is scored, what training changes, and what a downstream decision may infer. Its comparison unit is a learning target, rather than an individual paper or optimizer name. Representative methods make each contrast concrete; Appendix B supplies the complete source-level inventory.

Objectives: what signal should improve? External-event forecasting supplies an outcome that is held fixed while the report changes [17], [48], [49], [50]. Fixed-answer confidence training similarly isolates reporting [18]. Joint answer–confidence learning instead changes the answer distribution and the report together [51], [52], [19], [20], [53]. Reward-model interventions repair a preference signal or its interaction with expressed certainty [21]; expert guidance and consistency or verification feedback provide different targets for reports [23], [54], [55], [56], [22]. Ranking and relative-quality objectives prioritize useful ordering [24], [57], [25]. Numerical prediction distinguishes a conditional mean from a distribution of outcomes [26], [27]. Finally, action utilities and threshold-derived reporting scores connect feedback to a declared deployment loss [29], [28], [58]. Section 3 compares these families by the target and incentive each identifies.

Learning: how does the signal change parameters? Group centering and standardization determine how trajectory outcomes become credit [19], [30]. Token masks, trajectory baselines, paired comparisons, and process feedback localize that credit [31], [59], [60], [61]; direct report derivatives create a hybrid route when the reward itself depends differentiably on the parameters [26]. Supervised confidence fitting and post-hoc transformations provide simpler comparisons when answers or logits can remain fixed [32], [40], [62], [63], [33]. Online targets track a changing answer policy [34], [35], while distillation can align confidence with the information available to the student [64]. Finite-rollout objectives introduce an additional choice between scoring the realized empirical set and the law generating it [27], [16]. Section 4 studies these mechanisms and the conditions favoring each comparator.

Decisions: which property will be used? Selective answering and abstention connect ordering or numerical confidence to coverage and error costs [28], [65], [66], [25]. Adaptive computation requires success estimates available before additional work is purchased [37], [39], [36]. Class and structured outputs range from supervised label-token prediction to learned probability vectors [67], [22], [8], with guard models exposing sensitivity to the task’s safety labels and validation population [68]. Forecasting and judgment require either external-event probabilities, outcome distributions, or a stated annotator population [17], [69], [38], [27]. Section 5 develops these uses; Section 6 defines the corresponding comparisons. Jev supplies an industrial interface case, kept distinct from the public training algorithms [1].

Table 2 exposes three comparisons that recur throughout the survey. First, holding answers fixed isolates reporting, whereas joint training also changes which answers and outcomes are scored. Second, the reward determines whether training seeks probability magnitudes, useful ordering, or actions at a specified cost; a numerical output alone does not choose among these targets. Third, matching a mean, a probability vector, or a sampled predictive distribution requires different feedback and losses. These distinctions determine the appropriate baseline and evaluation: fixed-answer calibration, selective risk, cost-sensitive utility, or distributional scoring. They also explain why a shared PPO/GRPO optimizer or a similar output format is insufficient evidence that two methods solve the same statistical problem.

Table 2: What calibration-aware training learns. Families are compared by the object actually scored, the intervention, and the resulting decision semantics. Examples are representative; Appendix B retains all 49 source protocols and three controlled analyses.

Family / exemplars	Scored target and signal	Learning intervention	Decision use and interpretation
A. Probability reports: what event and information define the target?			
External events Forecasting [17], outcome-based RL [50]	Event probability; resolved outcomes with Brier/log rewards.	Update the report policy; the event definition stays fixed.	Forecast-based decisions; assess reliability and proper scores on held-out events.
Fixed-answer confidence Rewarding Doubt [18]	Supplied-answer correctness; clipped log-score reward.	Train confidence generation with the answer held as input.	Answer assessment; isolates reporting from the choice of answer during training.
Joint answer and confidence RLCR [19], Multi-RLCR [51], VPD [8]	Selected-answer confidence or candidate marginals; correctness plus report-error rewards.	Update the answer distribution and reports together.	Answering and selection; reporting and capability interact. A vector output may receive scalar supervision.
Prospective success CoCA [34], CALIBER [35]; capability probes [†] [36]	Policy success before the outcome; rollout-frequency targets.	Train report/answer segments, or fit a separate probe [†] .	Routing and compute allocation; estimates depend on the policy and information available when used.
Class probabilities Decision-token RL [31], RLVP [22]	Class-token probabilities or a generated vector; local class loss or teacher-guided reward.	Masked RL plus a local loss, or RL on generated vectors.	Classification; evaluate the full vector. Normalization or teacher alignment alone does not establish propriety.
Preference probabilities Judging with Confidence [38]	Annotator preference probability; Brier/log rewards on votes.	Train a probability-generating judgment policy.	Preference-sensitive judgment; the annotator population defines the event, rather than factual correctness.
Utility-derived reports Behaviorally calibrated RL [29]	Correctness confidence; a score derived from utility across thresholds.	Train confidence/value prediction; apply thresholds externally.	Cost-sensitive decisions; weighting the thresholds determines which reporting score is induced.

Reading the table. A fixed target is held fixed relative to the reporting intervention; it need not be observed without noise. A strictly proper score uniquely identifies a conditional probability when that probability is admissible. Composite rewards, restricted reports, changing policies, and finite optimization require separate analysis (Sections 3–4). [†] denotes a comparator without online rollout-reward training.

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Table 2 (continued). What calibration-aware training learns.

Family / exemplars	Scored target and signal	Learning intervention	Decision use and interpretation
B. Other learning targets and the role of simpler comparators			
Proxy / reader alignment RLMF [23], linguistic calibration [71]	Sampling-consistency-aligned confidence or reader belief; proxy/reader scores.	Change linguistic uncertainty and generated text.	Uncertainty communication; interpretation depends on the proxy or reader. Factual calibration needs separate evidence.
Relative ordering CAPO [24], RLSR [25]	Correctness order; pairwise or rank-weighted selective-risk rewards.	Change likelihood or confidence ordering.	Fixed-coverage selection; good ordering leaves the probability scale unidentified.
Selective actions TruthRL [28], UCPO [66]	Answer/abstain behavior; ternary utilities or group-dependent action credit.	Update the action policy under the chosen reward.	Selective answering; evaluate utility and coverage. Action likelihood is not correctness confidence.
Numerical means REAL [26]	Expected numerical rating; mean squared error plus likelihood.	Proposed policy and direct prediction gradients.	Point prediction; a correct mean does not identify a predictive distribution.
Predictive distributions DAR [27]	A finite set of generated outcomes; empirical CRPS.	Leave-one-out credit; group-standardized RL.	Distribution-sensitive decisions; distinguish the scored finite set from its generating law.
Supervised confidence [†] ConfTuner [32], answer assessment [63]	Fixed-answer correctness; semantic report error or binary prediction loss.	Directly fit confidence tokens or a separate estimator.	Answer assessment; tests whether rollout RL adds value beyond supervised reporting.
Post-hoc calibration [†] Temperature scaling [33]	Fixed-logit class probabilities; held-out likelihood loss.	Fit a scalar temperature with the generator fixed.	Probability rescaling; preserves class argmax and requires representative calibration data.

Cross-family comparison. Probability magnitudes, ordering, action utility, conditional means, and predictive distributions are different learning targets. Shared PPO/GRPO labels do not make them equivalent: group normalization, token masks, and direct report gradients change the implemented update (Section 4). † denotes a comparator without online rollout-reward training. The table states what an objective targets, not a guarantee of achieved calibration.

3. Objectives and joint capability

Calibration-aware learning draws on several forms of feedback: realized outcomes, answer verifiers, preference judgments, expert distributions, and the model’s own uncertainty signals. Figure 1 organizes these choices by what they reward. Proper scores provide a starting point for probability reporting; joint rewards also shape answer selection, while ranking and action objectives serve decisions that may require less than a complete probability model. Reward-model calibration adds a further intervention: changing the feedback that teaches the policy how to express confidence. This section connects these families through the outcome law and statistical quantity each objective is intended to learn.

3.1. Fixed outcomes, selected answers, and forecasting

For a fixed external event, changing a forecast ordinarily leaves the event itself unchanged. A strictly proper score uniquely favors its true conditional probability when that report is available and expectations exist [15]. Outcome-based forecasting develops this idea into a training pipeline. Future-as-Label uses resolved historical events with log-score rewards and restricts predictor inputs to pre-resolution information [49]. Outcome-based Reinforcement Learning to Predict the Future uses negative Brier rewards and compares alternative policy updates [50]. Proper-score forecasting then compares log, Brier, spherical, and Beta-family rewards under a shared training setup, showing how scores with the same truthful population optimum can produce different finite-budget learning profiles [17]. Together, these studies connect outcome supervision, temporal information control, reward geometry, and optimization.

A fixed-answer confidence task has a similar structure. Rewarding Doubt first generates an answer and then treats it as an input while training the confidence report with a log score [18]. The answer’s outcome law is held fixed during this reporting comparison. Joint answer-confidence training instead allows the policy to improve the answer as well as its report. RLCR addresses the resulting need to reward truthful uncertainty while retaining an incentive to answer correctly [19]. The distinction concerns the available intervention, rather than whether the output contains text.

Open-ended forecasting connects the two settings. OpenForecaster predicts a free-form future outcome and assigns a probability to that selected answer [48]. Although the world outcome is external, the proposition being scored depends on the model’s choice. Its free-form score is $S' = 2cZ - c^2 = Z - (c - Z)^2$, where Z indicates agreement with the resolved outcome. The final free-form reward adds another correctness term, whereas its fixed binary questions use negative Brier loss. Thus even within forecasting, a fixed-event probability and a selected-answer confidence call for different joint objectives.

Report timing also matters. CoCA and CALIBER distinguish a pre-generation estimate of policy success from a post-generation assessment using the realized answer and reasoning [34] [35]. Under a fixed policy, one binary outcome supplies valid squared-loss supervision for its conditional success probability; multiple independent outcomes reduce estimation variance. During joint training, the answering policy changes, and an early confidence token may itself become part of the answer context. Target construction must track these dependencies.

Throughout this section, p_π denotes a conditional success probability under policy π and a declared information set, and the report c is measurable with respect to that information. This probability may average over a randomized answering procedure or uncertainty remaining after information is withheld. Conditioning instead on a complete answer and a deterministic correctness rule makes the label zero or one. Population calibration given a report remains meaningful; a nondegenerate conditional target requires specifying which information or randomness is retained.

3.2. Separating capability, its distribution, and reporting regret

Let $S(c, z)$ be a proper score in reward orientation for binary z , with Bayes reward

$$G(p) = pS(p, 1) + (1 - p)S(p, 0).$$

Define its reporting regret by

$$D_G(p, c) = G(p) - \{pS(c, 1) + (1 - p)S(c, 0)\} \geq 0.$$

Under the standard regularity conditions, this is the regret associated with the convex representation of proper scores; strict propriety on a nondegenerate binary probability interval gives strictly convex G [15]. This classical geometry helps interpret what joint objectives value.

Consider $R = \alpha Z + \beta S(c, Z)$, where $\beta > 0$, and let $\mu = \mathbb{E}[p_\pi]$. A fixed report-independent additive constant shifts all policies equally and is omitted. For common input weights and finite expectations,

$$\begin{aligned} J(\pi, c) &= \alpha\mu + \beta G(\mu) \\ &\quad + \beta\{\mathbb{E}[G(p_\pi)] - G(\mu)\} \\ &\quad - \beta\mathbb{E}[D_G(p_\pi, c)]. \end{aligned}$$

The first line depends on mean capability. The second is a Jensen gap describing the distribution of conditional success probabilities. The last measures reporting error relative to the declared information. The identity applies to arbitrary admissible reports; truthful reports make its final term vanish. This reporting regret measures a different property from binned ECE, which aggregates errors within confidence bins.

For the correctness-plus-Brier reward used by RLCR,

$$R = Z - (c - Z)^2, \quad J = \mu^2 + \text{Var}(p_\pi) - \mathbb{E}[(p_\pi - c)^2].$$

Its calibrated conditional optimum is p^2 [19]. Increased reward can therefore reflect a higher mean success rate, a different success-rate profile, or more accurate reporting. The variance term concerns actual conditional success probabilities across information states. Randomizing reports around one unchanged probability instead increases reporting regret.

Different objectives can prefer different policies

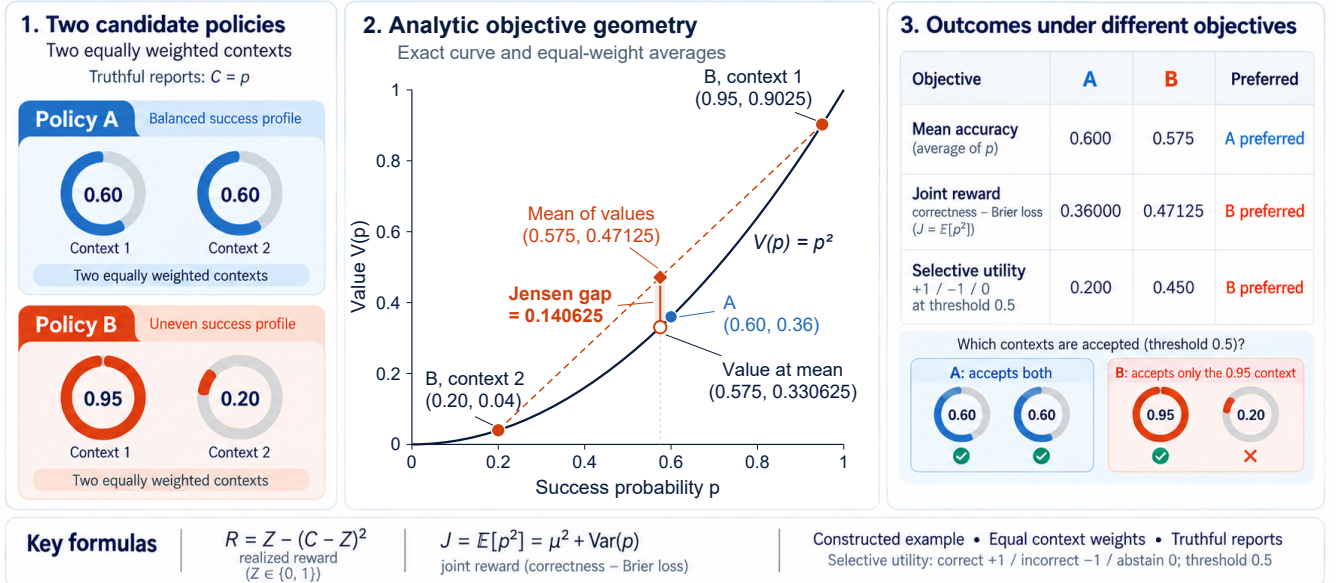


Figure 3: Classical proper-score geometry in an exact two-context construction. Left: policy A has success rates (0.60, 0.60); policy B has (0.95, 0.20); contexts have equal weight and reports are truthful, so reporting regret vanishes and $J = \mu^2 + \text{Var}(p) = \mathbb{E}[V(p)]$, with $V(p) = p^2$. Center: B’s expected value, 0.47125, is the midpoint of the chord joining its two contexts, exceeding $V(\mu_B) = 0.330625$ by the Jensen gap 0.140625; A’s contexts coincide at (0.60, 0.36). Right: A has higher mean accuracy, whereas B has higher correctness-minus-Brier-loss reward and higher selective utility at payoffs +1/-1/0 with threshold 0.5. The curves and marked coordinates are analytic; the policy illustrations are conceptual. The RLCR reward motivates the construction [19]; the geometry follows classical scoring theory [15].

Figure 3 illustrates the distinction with two equally likely contexts and a feasible policy family restricted to A and B. Policy A has success probabilities $(.60, .60)$; B has $(.95, .20)$. With truthful reports, B’s expected value is the midpoint of the chord joining its two contexts on the convex curve, and the Jensen gap above $V(\mu_B)$ (Figure 3, center) is what exact reward optimization rewards: it selects B although A has higher mean accuracy (Figure 3, right). The difference arises from the objective’s preference over a family that couples performance across contexts. Appendix A derives this illustrative construction and its assumptions.

The same change can be useful for selective decisions. If correct answers, incorrect answers, and abstention yield $+1, -1, 0$, respectively, truthful thresholding at $.5$ gives A expected utility $.20$. B answers only in the first context and obtains $.45$ (Figure 3, right, third row). A more uneven capability profile can therefore support more valuable selective decisions even when its average accuracy is lower. The deployment objective determines whether that allocation is beneficial.

This separates pointwise best effort from aggregate capability allocation. A nondecreasing calibrated value $V(p) = \alpha p + \beta G(p)$ favors greater reliability when it can be improved at one information state without sacrificing another. Across coupled policies, maximizing $\mathbb{E}[V(p_\pi)]$ also values their success-rate profiles. A finite affine correctness bonus changes the slope of V , while strict convexity remains. Appendix A gives the corresponding restricted-family ordering result.

Several conditions align the criteria. A pointwise-dominating feasible policy can optimize both when V is nondecreasing. With finite endpoint values and $p_\pi \in \{0, 1\}$, expected calibrated value is affine in mean accuracy, with ordering determined by its endpoint difference. Support on any two fixed probability values likewise reduces the comparison to an affine function of their mixture weight. When truthful reports are unattainable, policy-dependent reporting regret remains part of the comparison.

Accuracy weighting also gives a useful quantitative control. For bounded G , attainable truthful reports, and exact optimization over the same feasible family,

$$0 \leq \text{Acc}(\pi_{\text{acc}}) - \text{Acc}(\pi_\alpha) \leq \frac{\beta \text{osc}(G)}{\alpha}, \quad \alpha > 0.$$

Here π_{acc} maximizes accuracy and π_α maximizes the weighted reward. The bound in Appendix A concerns objective optima: increasing the accuracy weight limits their possible accuracy gap under the stated conditions. Neural optimization introduces the additional questions considered in Section 4.

3.3. Best-effort rewards and threshold-dependent utilities

Tan et al. describe joint confidence rewards through $f(c)$ for a correct answer and $g(c)$ for an incorrect one [20]. Their conditions combine proper reporting with a nondecreasing calibrated value over a specified probability interval. For differentiable rewards, its derivative is $f(p) - g(p)$. Their definition permits weak properness, including correctness-only rewards that retain an accuracy incentive while leaving the numerical report unidentified. This distinguishes best effort from the stronger requirement of eliciting a reusable probability.

For $R = Z - k(c - Z)^2$ with nonnegative calibration weight k , calibrated value is $(1 - k)p + kp^2$. On the full interval $(0, 1)$, $0 \leq k \leq 1$ preserves its nondecreasing slope. A log-score component has an outcome-specific score difference that becomes unbounded near an endpoint, placing a limit on what a fixed correctness bonus can guarantee globally [19] [20]. Log scoring remains proper for an external event or fixed answer; joint answer selection adds the best-effort requirement.

OpenForecaster provides a practical example of adjusting this balance [48]. Its final free-form reward, $Z + S' = 2Z - (c - Z)^2$, strengthens the reward for a correct low-confidence guess. In the reported validation comparison, near-zero-confidence “Unknown” responses fell from approximately 40% with S' alone to 4% after adding the correctness term. This supports accuracy weighting as a way to improve exploration in that setting, alongside the population-level role described above.

Decision-based derivations explain which costs a reward averages. The bounded answer/abstain utilities in [29] assign $+1$ or -1 to an answered outcome and $2t - 1$ to abstention at threshold t . Averaging these utilities under a uniform threshold density yields $2cZ - c^2$, exactly correctness-plus-Brier. Their transformation from the original utilities includes a scale depending on t , so uniform averaging is specific to that chosen representation. The unscaled

integral discussed in [20] instead produces $f(c) = c$, $g(c) = c + \log(1 - c)$. These are compatible constructions that weight cost regimes differently.

When one operating cost is stable, a direct action objective is also appropriate. TruthRL learns from rewards for correct answers, errors, and abstention [28]. Its intended outcome is a useful answering policy under the specified utility. Probability reporting offers a complementary interface when one model must serve multiple thresholds or changing costs, as developed in Section 5.

3.4. Ranking, proxy targets, and distributional outputs

Ranking objectives are well matched to candidate selection. CAPO orders correct and incorrect trajectories, while CONQORD aligns confidence differences with preference-model quality differences [24] [57]. Their useful output is a relative score; assigning that score a numerical probability interpretation requires an additional outcome-based calibration step. A generated confidence numeral can serve either role, so its objective provides the key distinction.

Proxy-based methods make feedback available when exact outcome supervision is costly. RLMP uses sampling consistency to supervise uncertainty expressions, and semantic-reward training derives confidence from relations among generated answers [23] [54]. Their transfer to correctness depends on the relationship between the proxy and the deployment outcome. The loss selects a statistical quantity as well: absolute deviation elicits a median, whereas squared loss elicits a mean, a relevant distinction for confidence-target methods such as DCPO [52].

RLVP extends expert guidance to complete class vectors. It rewards a verbalized distribution p with

$$R_{\text{Prob}} = \exp\{-(1 - \lambda)D_{\text{KL}}(e_Y \| p) - \lambda D_{\text{KL}}(q \| p)\},$$

where q is an expert’s soft distribution, e_Y is the observed one-hot label, and λ anneals from 1 to 0 [22]. Format and argmax-correctness rewards are added. Expert matching supplies fine-grained comparisons among generated vectors and can improve learned predictions during the curriculum. Its expected-score interpretation depends on the exponential transformation: at the hard-label endpoint, $R_{\text{Prob}} = p_Y$, so for a report issued before observing $Y \sim r$, expected reward is $r^\top p$, maximized by concentrating on a modal class. This analysis concerns the displayed probability component; teacher-correct filtering further conditions where it applies. Appendix A.7 separates the teacher-matching incentive, finite-training benefit, and endpoint scoring property.

Teacher information is another part of target design. CaOPD studies distillation from a teacher that receives privileged demonstrations or feedback unavailable to the deployed student [64]. It replaces confidence targets with the current student’s empirical success rate while retaining a token-level reverse-KL distillation objective. This is a calibration-oriented distillation comparator rather than a new proper-score RL reward. Its question-level success target illustrates why teacher certainty, student capability, and confidence conditional on a particular answer should be kept distinct. For both expert-guided RL and distillation, specifying the information behind a target clarifies what the student is being asked to predict.

Numerical tasks further distinguish functionals from distributions. REAL’s squared-error argument identifies a conditional mean and connects it to population Pearson optimality [26]. DAR instead scores an empirical distribution of numerical rollouts [27]. The mean supports squared-error decisions, while a fuller distribution can support quantile or tail-sensitive uses; Section 4 examines how finite sampling affects its score.

Finally, a probability target can belong to a reader. Linguistic Calibration rewards the log probability assigned to the correct answer by a surrogate reader of generated text, with a normalization penalty and policy KL [71]. LACIE uses correctness and listener acceptance to construct preferences [81]. These methods shape how uncertainty is communicated and used, complementing objectives that directly score a speaker’s numeric report.

3.5. Calibrating preference-derived reward signals

RLHF introduces a learned quality model between an outcome and the policy update. Leng et al. show that reward models can prefer a response with a higher appended confidence even when its answer text is held constant, including for rejected responses [21]. Their intervention targets this feedback channel: reward-model judgments should respond to confidence in relation to response quality, rather than treating confidence itself as an unconditional advantage.

PPO-M, calibrated reward modeling, fine-tunes an existing reward model using within-response comparisons. For a chosen answer, high confidence is preferred to low confidence; for a rejected answer, the preference is reversed. The resulting confidence–preference matching supplements the quality ranking learned by the original reward model. PPO then trains the answering policy using this modified scorer. This is a change to the reward model’s treatment of confidence, rather than a direct fit of a correctness posterior.

PPO-C, calibrated reward calculation, keeps the reward model fixed and first removes the explicit confidence report before scoring the answer. Writing its quality score as $\hat{r}$, reported confidence as c , and an exponential moving average of batch quality as Δ , the adjusted reward is

$$r = \hat{r} + w(\hat{r} - \Delta)(c - 0.5),$$

with $w = 2$ in the main recipe [21] (Eq. 3). Above-average judged quality rewards higher confidence; below-average quality rewards lower confidence. For fixed quality and center, the adjustment is linear in c , identifying confidence alignment with a quality proxy rather than an interior truthful-probability optimum.

The methods provide evidence that modifying preference feedback can improve subsequent factual confidence. On Llama3-8B with CoT on GSM8K, reported ECE is .2566 for PPO, .1909 for PPO-M, and .1546 for PPO-C [21] (Table 1). Their effects on ranking and broader instruction-following vary across settings. The family is therefore best understood as correcting a learnable bias in the reward signal, with probability calibration assessed on the resulting policy. It complements verifier-based proper scoring when preference data are the available supervision.

3.6. Intrinsic certainty and entropy as capability signals

A neighboring line uses confidence to improve reasoning itself. INTUITOR scores trajectories using average $D_{\text{KL}}(U||p_\theta)$, where U is uniform over the token vocabulary, and uses this self-certainty statistic as an intrinsic GRPO reward [82]. Positive-Advantage Reweighting instead regulates policy entropy by changing the contribution of positive-advantage tokens during correctness-reward training [83]. These mechanisms exploit or preserve properties of the generation policy, providing alternatives when gold-answer feedback is limited or exploration becomes concentrated.

Their uncertainty objects differ from a reported probability that an answer is correct. Self-certainty is a token-distribution statistic, and the entropy study’s confidence analysis compares likelihoods of correct and incorrect responses. Such signals can support capability learning or discrimination; calibrated risk decisions additionally require a validated mapping to outcomes. This boundary suggests a constructive division of labor: intrinsic signals can guide exploration and reasoning, while outcome-based scores or held-out calibration establish the numerical meaning of reports. The combined objective and its implemented update remain the units to evaluate.

4. Implemented learning

The literature implements confidence learning through several routes: trajectory rewards, token-level losses, external predictors, and combinations of these mechanisms. Baselines, normalization, token weighting, auxiliary gradients, and sampling procedures determine how the objective in Section 3 reaches the model. The landscape table groups these protocols by their targets and feedback; this section compares the updates within and across those groups. Trajectory learning is especially relevant when reasoning or decisions must change through sampled feedback, while direct fitting provides a strong starting point when the desired report loss is differentiable.

4.1. Centering and standardization preserve different information

Consider independently sampled reports for one prompt, scored against a shared external binary outcome. In the setting analyzed by Bereket and Leskovec, subtracting the mean reward of a group of size G gives

$$\mathbb{E}[\hat{A}_{\text{center}}(x, c)] = \frac{G-1}{G} A(x, c).$$

The proportionality is exact under the stated setup, before further transformations [30]. A conventional leave-one-out baseline removes the report’s contribution to its own baseline. Its justification relies on conditional independence of the other samples; coupled rewards, including scores shared across an ensemble, require a separate calculation.

Dividing by the group’s random standard deviation has a different effect. Reports within the group share a realized outcome, but their reward spread can differ according to whether that outcome is zero or one. The paper’s large-group approximation shows how standardization assigns outcome-specific inverse scales that depend on the current reporting policy. The update can therefore change the weighting responsible for proper elicitation. This approximation is not an exact characterization for every finite group or every proper score.

The authors’ CRISPR experiment reports ECE .292 with GRPO and .036 without the standard-deviation division [30] (Table 1). Its language-model launchers support a focused comparison by switching this normalization flag while retaining the other main settings. The appendix material separates that evidence from a discrepancy in the released synthetic notebook’s clipping path.

Other studies retain standardization and report improvements, including teacher-probability forecasting [70], separately credited capability and answer training [34], and decision-token training with an auxiliary direct loss [31]. Their targets and gradient paths differ from the stochastic-outcome setting. Conversely, RLCR’s mean-centered update accompanies a joint correctness/report reward, format incentives, and batch token normalization [19]. These comparisons motivate controlled estimator experiments with a common target, reward, and sampling budget.

4.2. Where feedback enters the update

Segmented and staged learning. DCPO and CoCA compute separate advantages for answer and confidence segments, with different target constructions: DCPO combines per-response correctness with a group success estimate, whereas CoCA predicts the current policy’s success before answering [52] [34]. Decision-token RL instead zeroes the final token’s policy advantage and adds a local classification loss [31]. CURE separates calibration preferences from a later factuality stage that masks confidence-related tokens [55]. These methods allocate feedback to particular token losses while retaining shared Transformer parameters. Their effect can be measured both on a fixed answer pool and on newly generated answers.

Fixing the answer as an input takes a different route. Rewarding Doubt samples and rewards confidence reports conditioned on an already generated answer [18]. Entropy-target RL trains a separate uncertainty adapter against a calibrated sampling-entropy target, while keeping answer generation fixed [74]. The latter amortizes an offline sampling-and-calibration pipeline; its target is question-level error under that pipeline, even though the reporter also sees a particular answer. These designs help isolate assessment from answer selection and expose the data and sampling costs of producing confidence supervision.

Sequence and group credit. C²GSPG couples a sequence policy objective to binary cross-entropy on the geometric mean of token probabilities. Its advantage divides the centered reward by one minus the old sequence score, introducing confidence-dependent weighting without group-standard-deviation division [61]. UCPO instead separates correct, incorrect, and abstaining outputs, adapts the abstention advantage to group composition, and filters groups lacking the required outcome types [66]. The first trains an internal likelihood score as a correctness proxy; the second trains discrete uncertain behavior. Their update rules respond to different objects despite both using the language of confidence calibration.

Paired and process feedback. RAC distributes confidence-ranking and clean-versus-corrupted evidence penalties across paired multimodal rollouts [60]. RLCM fits a probe to the success rate of answers forced from intermediate reasoning prefixes; the probe receives a supervised loss, while the policy receives final correctness plus an aggregate confidence-margin reward [59]. RefGRPO adds a decaying bonus for a binary self-assessment made after environment feedback [73]. Together, these methods extend credit beyond a final numerical report: to evidence perturbations, reasoning prefixes, and post-action reflection. Their relevant evaluations include directional behavior, prefix success prediction, and reflection accuracy, respectively.

Feedback and target adaptation. PPO-M first retrains an existing preference reward model to rank confidence differently for chosen and rejected responses. PPO-C instead removes the explicit confidence number before reward evaluation and adjusts the resulting quality score according to confidence and a moving reward baseline [21]. This addresses the feedback source itself. Success estimates in CoCA and CALIBER also evolve with the answering policy [34] [35]. Refreshing such targets keeps them aligned with current behavior, while adding sampling noise and dependence when the same rollouts supply both targets and advantages. CALIBER’s warmup illustrates a further issue: a statistically meaningful group target can have too little within-group variation to compete with other rewards.

Finally, reward magnitude interacts with KL penalties, format bonuses, and clipping. A positive affine transformation preserves a fixed report optimum but can change its weight against an unchanged auxiliary term. Token aggregation creates another weighting choice. DAPO weights tokens using the observed total response-token count, while Dr. GRPO replaces response-length and question-level reward-standard-deviation normalization with fixed normalization choices [84] [85]. The two methods address related length-weighting issues through different estimators. Active-token reductions used in calibration protocols add further implementation choices [19] [31]. A reproducible description therefore includes the baseline, scale, mask, auxiliary losses, and reduction rule, in addition to the optimizer’s name.

4.3. Differentiable rewards require a direct prediction gradient

Policy dependence has a second, analytically distinct meaning. A reward may depend differentiably on the current parameters even after a trajectory has been sampled. If

$$J(\theta) = \mathbb{E}_{\tau \sim \pi_{\theta}}[r_{\theta}(\tau)],$$

then, under the usual conditions for differentiating an expectation,

$$\nabla_{\theta} J = \mathbb{E}[r_{\theta}(\tau) \nabla_{\theta} \log \pi_{\theta}(\tau) + \nabla_{\theta} r_{\theta}(\tau)].$$

The first term changes the trajectory distribution; the second changes the evaluated prediction on that trajectory. Treating every reward as an external stopped-gradient scalar omits the latter.

REAL makes this distinction explicit [26]. Its numerical prediction is a probability-weighted sum of score-token values. The proposed reward combines negative squared error with an auxiliary target-token log likelihood, so the prediction term can backpropagate through both components. A parsed numerical string instead receives a score through its discrete generation action; periodically refreshed success labels introduce yet another kind of policy dependence.

The exact identity and the practical algorithm remain separate. REAL’s paper standardizes and clips leave-one-out advantages, and weights the direct prediction gradient by β . This prediction weight is separate from reference-policy KL regularization; even $\beta = 1$ retains the transformations of the trajectory term. Its SFT-initialized ablation reports Pearson $r = .679$ and RMSE .968 at $\beta = .01$, versus $r = .667$ and RMSE .935 at $\beta = 1$ [26] (Table 9). Correlation and numerical error favor different settings, motivating evaluation matched to the desired regression statistic.

The public implementation also warrants version-qualified reporting. At the audited commit, the example launchers set the direct-gradient and dedicated KL coefficients to zero; the specialized actor separately normalizes negative-MSE and target-probability rewards, rather than the paper’s combined log-likelihood reward. Its GPG loss uses an unclipped score-function term, although advantages are clipped. These observations identify differences between the paper formulation and a public execution path. They do not reveal which settings produced the published checkpoints or ablation runs. The appendix material supplies fixed-commit evidence and keeps those conclusions separate.

4.4. When does reinforcement learning add a distinct intervention?

Supervised estimation, prompting, and post-hoc fitting provide substantive alternatives to policy learning. Early verbalized-confidence SFT learns confidence words or numbers from empirical subtask accuracies [40]. Kapoor et al. compare supervised probes and language-formulated correctness classifiers on fixed generated answers [63]. Tian et al. elicit confidence from existing RLHF models through prompts, with optional labeled temperature fitting [41]; Xiong et al. systematically compare prompting, sampling, and aggregation choices [76]. These approaches separate the cost of obtaining a useful confidence signal from the cost of changing the answering policy.

ConfTuner offers a particularly direct training comparator. It constructs a distribution q_i over tokens denoting numerical probabilities i/N , and minimizes

$$\ell(q, z) = \sum_{i=0}^N q_i (z - i/N)^2.$$

It obtains q_i from a softmax restricted to candidate confidence tokens and differentiates this expectation directly [32]. Summing over the finite reporting support removes Monte Carlo variance from sampling the confidence action. The categorical q represents alternative reports; their numerical values carry the intended correctness meaning. Label noise, the reporting grid, and distribution shift remain relevant.

An external predictor can also leave generation unchanged. Generalized Correctness Models learn from labeled histories across generators, using the question, response, and generator identity [79]. Their information and label requirements differ from an internal confidence head. At the post-hoc end, temperature scaling fits held-out outcomes [33] [86], while Dual-Align fits a temperature using a pre-trained model’s final distribution and layer signals as an unlabeled reference [80]. Choosing between these comparators depends on access to logits, model internals, target-domain labels, and a suitable reference model.

For class probabilities, answer-only CFT and its regularized variant provide supervised comparisons [77]. Label smoothing changes the SFT target over the vocabulary [62]; CATTO augments DPO with a confidence-related token surrogate [78]. These methods address probability scale through different training signals. CATTO’s surrogate is linear in confidence for a fixed soft target, so its empirical classification results should be compared without importing a strict-propriety guarantee. BinaryPPO represents a neighboring approach that treats binary labels as actions [75]; its reported classification results motivate a matched supervised baseline, while unresolved objective signs and limited probability-quality evaluation place it outside the established calibration evidence.

Table 3: Starting comparisons determined by the intended intervention and available gradients.

Intended intervention	Justified starting comparator	What makes RL a distinct choice?
Correct probabilities from frozen scores	Held-out fitting [33] [86]; reference-model fitting where labels are unavailable [80]	Changing the generator or information acquired, beyond remapping a score
Assess fixed answers	Prompted/sampled reports [41] [76]; supervised estimation [40] [63]; exact expected-score training [32]	Exploring an assessment process whose consequences require sampled feedback [18]
Add a correctness predictor without changing generation	Historical cross-model prediction [79]	Adapting the answer policy or learning coupled behavior rather than only a separate estimator
Learn calibrated class outputs	Answer-only CFT [77], label smoothing [62], class-logit supervision [67]	Changing sampled reasoning together with classification feedback [31] [22]
Improve answers and report their reliability	Matched supervised answer/confidence training; preference regularization [78]	Updating sampled reasoning and answer choices from verifier or calibrated reward-model feedback [19] [21]
Improve a differentiable numerical prediction after reasoning	Regression-aware supervised training	Learning which reasoning trajectories to sample while retaining prediction gradients [26]
Optimize ordering or one-cost acceptance	A rank or action objective with matched supervision	Improving sequential choices from feedback, without requiring probability identification [24] [28]
Learn a rollout-induced predictive distribution	A distributional supervised baseline	Changing the generation distribution through sample-set feedback [27]

These starting points form a comparison design, rather than a ranking of algorithms. Figure 4 summarizes the three protocol families by the gradient routes they use: a numeral parsed after generation receives only the score-function route (RLCR [19]); a fixed answer with an exactly differentiated report loss uses only the direct route (ConfTuner [32]); REAL’s ideal hybrid formulation makes both routes available [26]. Direct fitting is attractive when the desired expectation is tractable; RL is useful when reasoning, retrieval, answering, or abstention must adapt through

outcome feedback. Hybrid methods retain both gradients when available. The strongest evidence for a distinct RL contribution identifies the behavior being changed, supplies the relevant supervised or post-hoc baseline, and matches access, labels, and compute.

Two routes from reward to learning

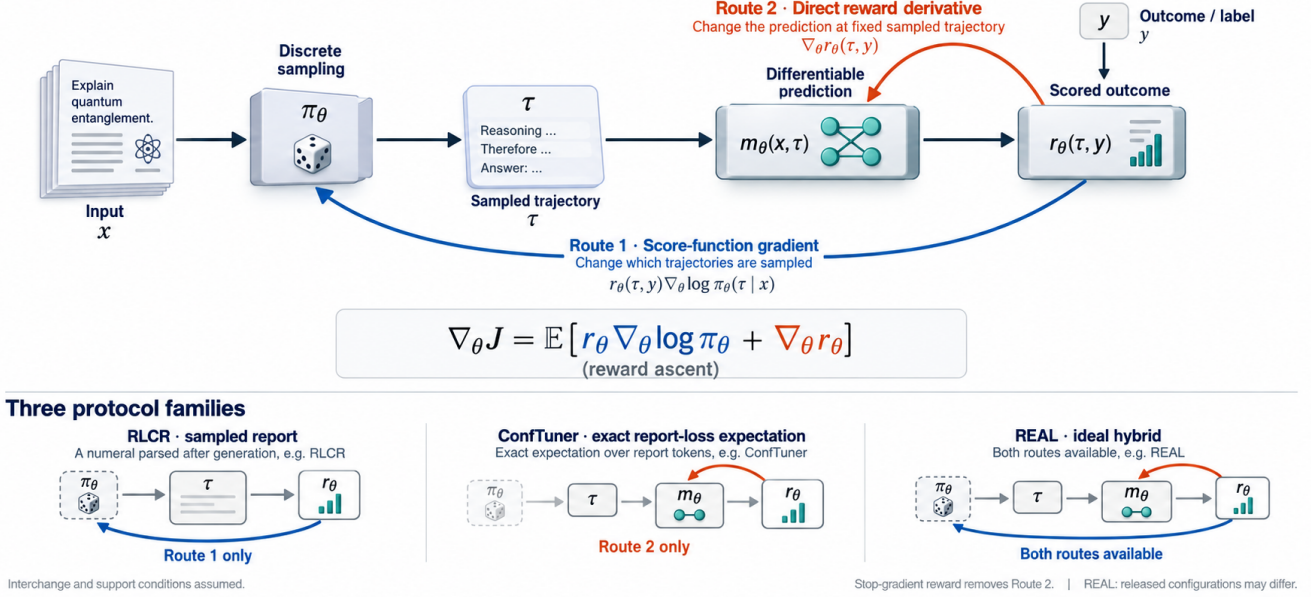


Figure 4: Two routes from reward to learning. Top: a sampled trajectory $\tau \sim \pi_\theta$ conditions a differentiable prediction $m_\theta(x, \tau)$, scored against an outcome y . Under the usual interchange and support conditions, the expectation gradient contains a score-function term that changes which trajectories are sampled (route 1) and a direct reward derivative that changes the prediction on a sampled trajectory (route 2). Stopping the reward gradient removes route 2. Bottom: a numeral parsed after generation (e.g. RLCR [19]) uses route 1 only; a fixed answer with an exactly differentiated report loss (e.g. ConfTuner [32]) uses route 2 only; REAL motivates the ideal hybrid identity with both [26]. Stabilization and the audited public REAL launcher are separate from this identity; the diagram does not assert that every released run used both routes.

4.5. Finite ensembles change the scoring problem

Scoring several numerical rollouts introduces a distinction even before optimization. A proper score for an issued empirical distribution need not be a fair score for the distribution that generated its members. “Fair” here has the ensemble-forecast meaning formalized by Ferro, rather than a demographic interpretation [16].

For independent $X_1, \dots, X_K \sim F$, a fixed observation y , and finite first moments, the usual empirical CRPS is

$$C_K = \frac{1}{K} \sum_i |X_i - y| - \frac{1}{2K^2} \sum_{i,j} |X_i - X_j|.$$

Its expectation satisfies

$$\mathbb{E}[C_K] = \text{CRPS}(F, y) + \frac{\mathbb{E}|X - X'|}{2K}.$$

The extra term penalizes dispersion of the sampling distribution. Replacing K^2 in the pairwise denominator by $K(K-1)$, for $K > 1$, removes this term under independent sampling [16]. This classical adjustment illustrates why rollout count can change both the expected training criterion and its estimator variance. Figure 5 shows the size of the effect in the Gaussian example of Appendix A: for $K = 2, 4, 8$ the conventional empirical score is minimized at an under-dispersed spread, whereas the fair score is minimized at the true spread for every $K > 1$.

DAR uses the empirical K^2 score and assigns rollout i a leave-one-out contribution $r_i = C_{K-1, -i} - C_K$ [27]. Its public reward manager implements those denominators. The resulting scalar rewards then undergo GRPO group centering and standardization. The effective K excludes invalid numerical parses, with fallback and penalty rules for exceptional groups. Defining a set score, assigning credit, and transforming the update are therefore three separate

Small ensembles can reward forecasts that are too narrow

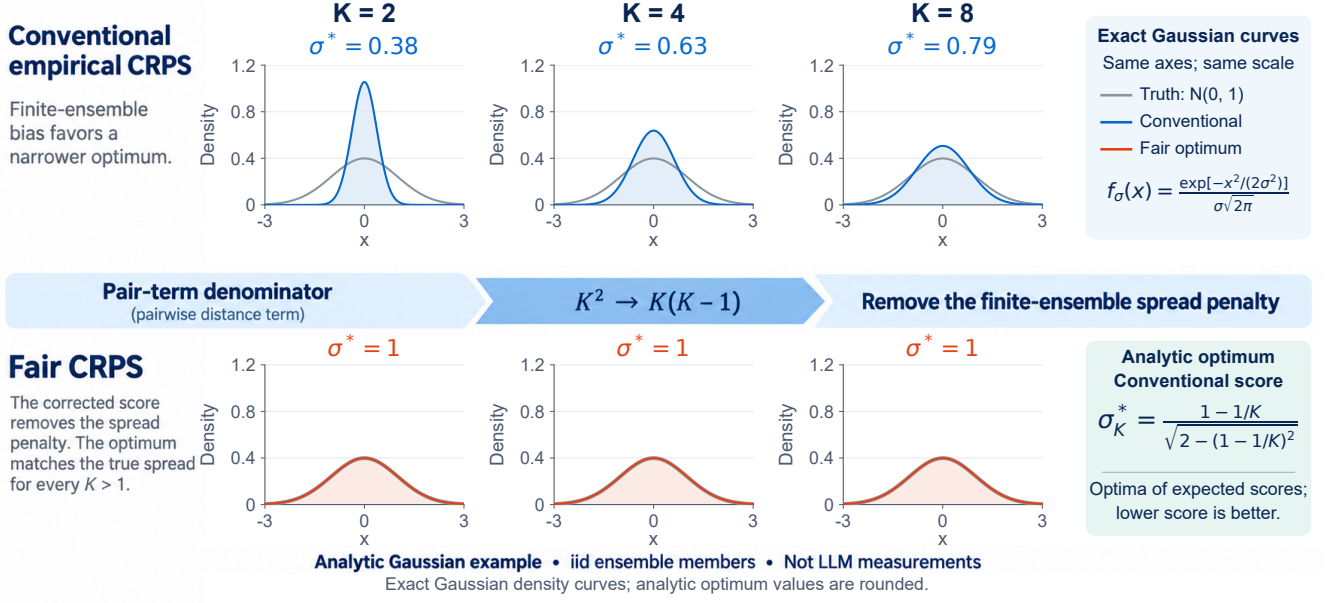


Figure 5: Classical finite-ensemble scoring example [16], computed from the analytic formula in Appendix A. Truth is $N(0, 1)$; the forecast generator is $N(0, \sigma^2)$ with K iid members. Top: the densities at the conventional empirical-CRPS optima are under-dispersed: $\sigma_K^* = a/\sqrt{2 - a^2}$, where $a = 1 - 1/K$, giving approximately 0.38, 0.63 and 0.79 for $K = 2, 4, 8$. Bottom: replacing the pairwise denominator K^2 by $K(K-1)$ yields the fair score, whose expected optimum is $\sigma^* = 1$ for every $K > 1$, so forecast and truth coincide. The curves are exact Gaussian densities, not LLM measurements. The iid correction alone does not characterize DAR’s filtered, normalized update [27].

operations. The leave-one-out contribution retains the empirical-score denominator rather than implementing the fair-score correction.

The identity motivates a controlled test varying the pairwise denominator and rollout count while holding sampling, validity handling, and advantage transformations fixed. Dependent generations, deterministic quantiles, and filtering require their own assumptions. DAR’s reported CRPS improves over its MSE comparator, while SFT remains better on KBSS [27] (Table 5). This mixed result makes the intended distributional quantity and the actual estimator useful organizing principles for comparing regression-oriented RL.

5. Decisions and interfaces

The decision determines which statistical property is useful. A service that must answer or abstain under one fixed cost, an auditor that reviews a fixed quota, and a controller that allocates another generation have different requirements. This section connects those requirements to method choice. The relevant interface can be a label, a scalar, a vector, or a generated report; its shape alone does not establish what training has learned.

5.1. Fixed costs and reusable probabilities

Suppose accepting an answer pays $b > 0$ when correct and $-\ell$ otherwise, while rejection pays zero. For a score C calibrated for correctness Z ,

$$\mathbb{E}[u(\text{accept}, Z) \mid C = c] = (b + \ell)c - \ell, \quad \tau = \frac{\ell}{b + \ell}.$$

Acceptance at $c \geq \tau$ is therefore optimal among policies observing only that score, with either action optimal at equality. The qualification is consequential: this uses fixed payoffs and calibration under the deployment joint distribution. A policy with additional input information may do better; arbitrary contextual costs require that

information to enter the decision. Section 3 and the mathematical appendix distinguish this result from guarantees about the trained generator.

Probabilities are particularly useful when the cost ratio changes after training. The same report can support different thresholds for different services, provided its event and population remain appropriate. A score with good ordering but unknown probability semantics can support a threshold fitted on validation data; it cannot directly inherit a cost-derived numerical threshold. Behaviorally calibrated RL implements the reporting approach explicitly: its successful confidence variant generates a report and applies the answering threshold externally [29].

When the objective instead fixes the fraction of answers to accept, only ordering and tie handling determine the selected set. Under common payoffs, ordering by conditional success is the appropriate target, and a selective-prediction objective is a useful comparator. RLSR’s HotPotQA-trained in-domain comparison illustrates the distinction: relative to RLCR, AURC improves from .51 to .44 and accuracy at 10% coverage from 53.5% to 66.0%, while ECE worsens from .05 to .24 [25] (Table 1). These results motivate choosing and evaluating the desired operating objective before choosing a training loss. Safety detection illustrates the same need at the tail of the score distribution: Reasoning’s Razor reports settings where reasoning improves aggregate accuracy but reduces recall at a strict false-positive-rate threshold [87]. Such a result motivates reporting the relevant operating region alongside average calibration and discrimination.

5.2. The same ordering can support different transported thresholds

Consider four equally likely contexts with correctness probabilities (.1, .4, .6, .9). Compare truthful reports with their monotone sharpening (.001, .1, .9, .999) and the constant base-rate report .5. Acceptance pays +1 for correctness and -4 for error, so the probability threshold is .8. The following are exact population calculations, rounded for display. The full numerical table is retained in Appendix A.

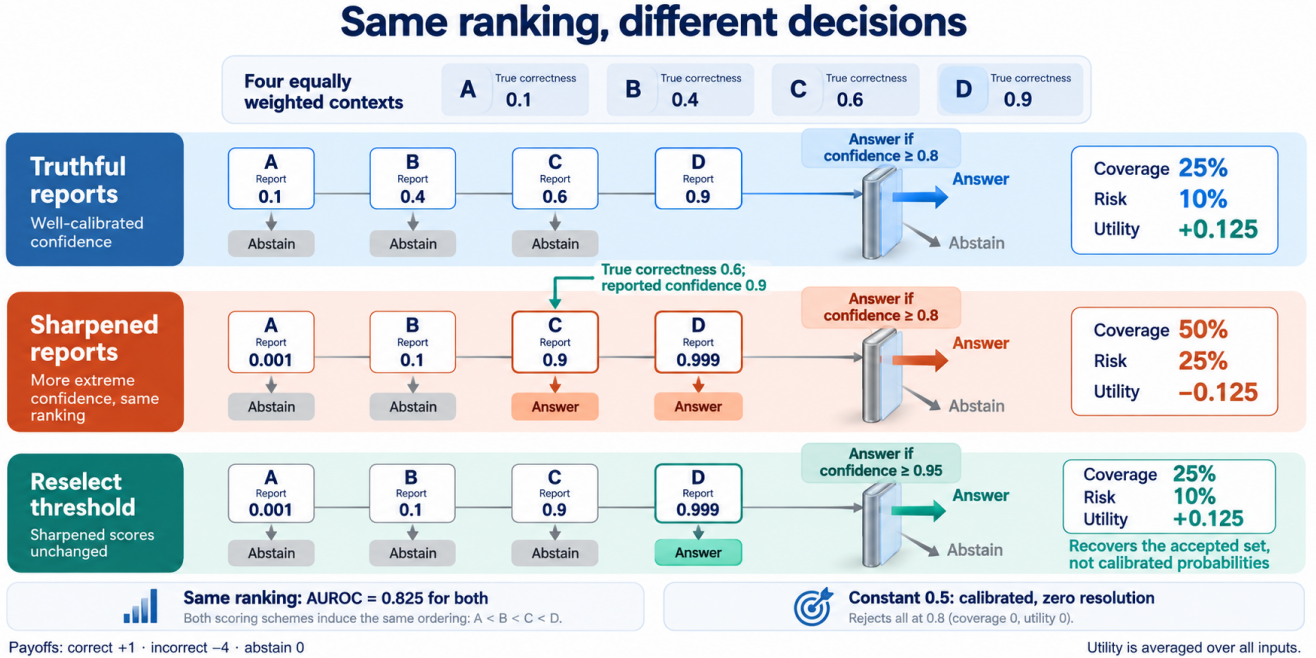


Figure 6: Exact four-context decision example with equal weights and correctness rates (0.1, 0.4, 0.6, 0.9). At the cost-derived threshold $\tau = 0.8$, truthful reports accept only context D, giving coverage 25%, risk 10%, and expected utility +0.125 per input under payoffs +1/-4/0. Monotone sharpening to (0.001, 0.1, 0.9, 0.999) preserves ranking but also accepts context C, giving coverage 50%, risk 25%, and utility -0.125. Reselecting the sharpened-score threshold to 0.95 recovers the original accepted set and utility; this is threshold adaptation, not recalibration. Risk is the error rate among accepted answers; utility is averaged over all inputs. Both reporters have AUROC 0.825. The constant 0.5 report is calibrated with zero resolution and rejects all inputs at 0.8. Finite validation data determine how reliably an alternative threshold can be selected. Full metrics appear in Appendix A.

For the metric comparison in Appendix A, reliability means $\mathbb{E}[\mathbb{E}[Z \mid C] - C]$, rather than the squared reliability term in the Brier decomposition [42]. Sharpening preserves ranking but moves the .6-success context across the transported .8 threshold and turns positive utility into negative utility (Figure 6). The constant report is calibrated, yet supplies no resolution between the contexts; rejecting everything is optimal given that report alone.

The truthful and sharpened reporters nevertheless have identical attainable threshold-based ROC and risk-coverage frontiers (Appendix A; Figure 6 illustrates the corresponding accepted sets). A sharpened-score threshold in (.9, .999] recovers the truthful accepted set. Independent validation can select a replacement threshold, although finite samples do not guarantee recovery. The example establishes the value of probability semantics for transporting thresholds, while retaining the option of learning thresholds from labeled validation data.

The invariance follows from a strictly increasing transformation of the same scalar score, fixed answers and outcomes, and consistent treatment of ties. Equal AUROC values alone do not imply identical curves. Nor does this argument apply automatically to maps that merge ties, a changed answer policy, or general multiclass temperature scaling followed by maximum-confidence readout. A useful experiment therefore reports both cost-derived thresholds and independently selected thresholds: their difference reveals whether an improvement comes from numerical probability semantics or from the attainable selection frontier.

5.3. Learning the action when the operating objective is fixed

For a stable answer-or-abstain service, learning the desired action can be sufficient. TruthRL assigns +1, 0, and -1 to correct answers, abstentions, and incorrect answers [28]. For a fixed candidate with success probability p , answering has expected reward $2p - 1$, giving the ideal threshold .5. The training feedback identifies which action is desirable without uniquely identifying a numerical probability. This makes direct-action RL a relevant baseline when deployment uses that same utility and retraining can accommodate changes.

Its performance should be judged through both errors and retained correct answers. On CRAG with retrieval, Llama3.1-8B-Instruct trained with TruthRL reports accuracy 56.6% and hallucination 19.4%, versus 60.3% and 39.5% for binary-reward RL [28] (Table 1). These rates use all questions as their denominator. The result shows a useful change in operating behavior, including fewer correct answers relative to that comparator; it does not measure calibration of an unreported probability.

Action learning also introduces an optimization question. The abstention-collapse analysis studies a shared, bounded answer gate under error-penalized rewards [65]. In that setting, both the task gradient and the policy-to-reference KL gradient contain a saturation factor; a predominantly low-success training population can suppress answering even on easier inputs. Its controlled gate and constructed QA-mixture experiments motivate comparing direct abstention with mandatory-answer probability training followed by an external threshold. The analysis relies on the shared parameterization and stated training conditions, so it complements TruthRL’s operating-point evidence rather than predicting collapse for every answer-or-abstain policy.

The complementary reporting strategy is attractive when one model serves multiple cost regimes or when operators need to adjust coverage without retraining. Evaluation incentives matter here as well: *Why Language Models Hallucinate* explains how binary correct/incorrect grading can reward guessing relative to abstention, and motivates grading rules with an explicit confidence threshold [88]. This connects the training payoff to the behavior a benchmark will select. For an existing generator, supervised correctness estimation or post-hoc calibration supplies a natural starting comparator. RL adds a distinct intervention when the intended change includes the generated answer, reasoning, or abstention behavior itself, as discussed in Section 4. The choice is about the behavior to change and the information deployment needs.

5.4. Adaptive computation depends on when the score is available

Deciding whether to accept an existing answer differs from deciding whether another generation is worth its cost. The P(IK) formulation [39] already separates pre-answer capability from P(True) evaluation of a candidate. Capability confidence concerns the expected success of a specified stochastic decoder on a query; response confidence concerns an already sampled answer [36]. A pre-generation router needs the former target for each available model, whereas a post-generation acceptance gate needs evidence about the actual candidate. A score acquired after execution or external verification has still different information and should be assessed at that stage.

Under independent trials with the same query-level success probability p , the probability of at least one success after k samples and the marginal gain from another sample are

$$P(\text{success by } k) = 1 - (1 - p)^k, \quad \Delta_k = p(1 - p)^k.$$

Probability magnitudes enter this allocation rule [36]. With a value assigned to success and a sampling cost, they support allocating computation where the expected gain is largest. The same principle suggests comparing model-specific expected benefits and costs for routing, rather than routing from the concentration of an unrelated label distribution.

The assumptions delimit the supported use. The decoder must remain the one whose capability was estimated; correlated or adaptively changed samples need a different gain model. Moreover, success-at- k assumes that a successful candidate can be recognized, or measures an oracle event. It is not automatically the performance of the deployed selector. The capability-prediction study reports Qwen3-8B pass@64 simulation MSE .0205 for its capability probe versus .1486 for an oracle response-calibration comparator on MATH-500 [36] (Table 3). This is a useful target-matching result, not evidence that capability scores should replace response confidence when accepting an answer. Routing and allocation evaluations should record the information available before each decision and charge for obtaining it.

5.5. Richer interfaces must expose the object the decision uses

Direct decision interfaces have several learning routes. SALSA maps classes to existing label tokens and fits their renormalized logits by supervised cross-entropy [67]. GLiClass uses encoder-based text-label representations, including independent sigmoid scores for multilabel tasks, with a PPO stage in its broader training pipeline [89]. BinaryPPO frames binary label prediction as policy optimization [75]. These systems supply useful architecture and training comparisons; their classification results motivate separate proper-score and reliability evaluation when scores will be consumed as probabilities. In guard models, label-token readouts are already operational, and post-hoc calibration depends on the calibration-set distribution [68].

An output vector is valuable when downstream costs distinguish among several labels. Its training target still requires inspection. Verbalized Probability Distribution emits candidate probabilities and selects the maximum, but inherits a scalar correctness-plus-Brier reward for that selected answer, together with a penalty for an invalid probability sum [8]. It nonetheless reports improved full-vector Brier scores after RLCR training: .387 to .347 on MedQA for Qwen3-4B-Instruct [8] (Table 6). This is evidence that scalar supervision can improve the richer output empirically; it does not identify every nonselected probability through the reward. A downstream cost matrix using the full vector therefore deserves full-vector evaluation.

Inference alone can also preserve useful information: aggregating a judge’s rating-token distribution instead of taking its modal score improves the evaluated judgments in many settings [69]. This provides a comparator for learning a new judgment distribution. Preference judgments introduce another distinction. Judging with Confidence targets the probability that a specified annotator population prefers B to A, using Brier or log rewards on binary observations [38]. Its main population is persona-conditioned Gemini. Evaluation compares soft preference proportions using MSE, but binarizes them into majority labels and removes ties for ECE and Brier [38]. For Gemma, RL-Brier improves soft-target MSE over temperature scaling (.0764 versus .0839), while temperature scaling has lower reported ECE (.0827 versus .0879) [38] (Table 1). Modeling disagreement and predicting the majority winner are useful, distinct tasks; the evaluation should match the intended one.

For numerical decisions with finite second moments, squared-error loss selects a conditional mean, whereas quantile or tail-sensitive decisions need other distributional features. REAL trains an expected numerical judgment using regression feedback; DAR scores an empirical distribution of numeric rollouts [26], [27]. Their optimizer and finite-ensemble issues belong in Section 4. The decision-level lesson is to expose and evaluate the statistic that the loss actually needs, then retain a fuller predictive object when it supports additional uses.

RLVP supplies a complementary full-vector route: it generates reasoning and JSON class probabilities, then trains their exponentiated-KL agreement with an XGBoost expert and observed labels [22]. This changes both the output interface and the feedback available to the learner. Its teacher is trained on each task’s training split and predicts soft targets on that same split, with teacher-correct filtering; this is neither an independently observed

posterior nor evidence of test-label leakage. Matched supervised hard/soft-target comparators are therefore useful, as are cross-fitted teacher targets and a comparison with untransformed log-score rewards. The method remains autoregressive text generation internally, so its machine-readable output should not be described as a token-free decision head. Its CoT is generated alongside the probabilities; assessing faithfulness calls for a separate reasoning evaluation. Calibration-aware on-policy distillation offers a related information-set comparison [64]: a teacher can see privileged demonstrations or feedback while the deployed student sees less information. Its rollout-based student confidence targets help align the report with the student’s own success rate. RLVP’s training-set expert and CaOPD’s privileged-context teacher are different protocols, but both motivate recording teacher access, target construction, and the deployment information set.

5.6. Jev as an industrial interface case

TypeSafe presents Jev as a system for typed decisions and probability outputs, generated in parallel for use inside software [1]. Its official expansion is **Reinforcement Learning for Calibrated Decisions** [90]. Classification, routing, and extraction are plausible beneficiaries: applications can consume bounded outputs directly and retain probabilities for later cost-sensitive decisions. The public materials establish that design direction, while leaving the full training recipe insufficiently specified for identification with an open academic algorithm.

The API also illustrates a semantic issue that matters at deployment. Its documentation distinguishes a probability vector from an additional confidence statistic derived from distribution shape [91]. Before applying the cost threshold in Section 5.1, an application must establish that its chosen field estimates the relevant event. A useful default statistic can support validation-based gating without thereby becoming a correctness probability.

Community artifacts implement different parts of this design. The Qwen-branded readout engine, `jev-on-a-laptop`, and JEVfire explore constrained candidate scoring and output assembly using existing models [92], [93], [94]. `AlexWortega/openjev` instead describes a supervised sequence classifier trained with cross-entropy on entailment-style labels [95]. These provide practical comparators for interface efficiency and classifier training; their names do not establish a reproduction of TypeSafe’s RL method. Appendix D compares the eight inspected industrial sources and the controls needed to measure their benefits.

For multistep structured outputs, grammar-constrained decoding adds another distributional choice. Park et al. distinguish local token masking from sampling the language model conditioned on eventual satisfaction of the grammar [96]. The latter depends on future valid-continuation mass, and their ASAp procedure approaches that conditional distribution through additional sampling. This result concerns a sequence distribution conditional on a grammar, rather than correctness calibration. A single-step softmax over fixed candidate labels is a different case; its numerical meaning depends on the candidate set and outcome supervision. The productive systems question is which combination of training, probability readout, and serving design improves the application’s measured decisions.

6. Evidence and evaluation

An evaluation should establish what improved, under which information and resource constraints, and for which decision. Calibration-aware training may change answers, confidence reports, parsing success, and the subset of queries a system accepts. A single aggregate metric cannot identify which change produced the gain. We propose an evaluation contract that separates these effects while retaining the native system’s end-to-end performance.

6.1. Separate reporting quality from policy improvement

Two complementary evaluations are needed. A **fixed-answer evaluation** compares reporters on a common collection of questions, answers, histories, and labels. It measures how well each reporter assesses the same outcomes. A **native-policy evaluation** lets each trained system generate its own answers and confidence, then measures predictive quality and decision utility under its intended deployment procedure. The first isolates a useful comparison; the second preserves the benefit of generating better answers. Neither substitutes for the other.

Where interfaces permit, a crossed diagnostic adds two intermediate comparisons. Let π_0, π_1 be the original and updated answer policies and R_0, R_1 their confidence reporters. Generate answer pools from each π_i , and evaluate

$$C_{ij} = R_j(X, A_i, H_i), \quad i, j \in \{0, 1\},$$

against labels for the same A_i . Comparing reporters within a pool tests reporting changes on fixed answers; comparing pools reveals dependence on the answer distribution. The diagonal entries should also be compared with native operation, because an imposed reporting interface may itself change behavior.

The diagnostic is interpretable when the reporters have compatible inputs. A reporter may require unavailable hidden states, a particular reasoning format, or information that another policy never produces. Crossed results should identify those compatibility restrictions and any out-of-distribution use. Outcome feedback and reference-answer access should be matched, and reported uncertainty should be measured before the decision it informs.

6.2. A common evaluation contract

The following contract makes the comparisons executable. Its purpose is to align evidence with the intended claim, rather than require every task to optimize the same metric.

Table 4: A common contract for interpretable evaluation.

Evaluation component	Required comparison or record
Target and information	Specify the event, label rule, report timing, and evidence visible to the reporter; retain these fields for each example
Answer and reporter changes	Report fixed-answer and native-policy results; add compatible crossed evaluations when diagnosing their interaction
Supervision and selection	Record labeled inputs, labels per input, calibration data, checkpoint/threshold selection data, and an untouched final test population
Inference resources	Count reasoning tokens, candidate samples, retries, tool calls, and auxiliary judges; compare fixed-budget performance or explicit cost-quality curves
Output validity	Report attempted and valid counts, parsing/fallback rules, and utility of failures; show valid-only statistics as a separately labeled conditional result
Prediction and decisions	Pair the target’s proper loss and calibration diagnostics with ranking, coverage, and utility at declared operating points
Uncertainty of estimates	Give sample counts, accepted/error counts, paired uncertainty intervals, and seed variability where training randomness is material

Holdout design should follow the selection procedure. Fit transformations and choose checkpoints or thresholds without consulting the final evaluation labels. If a benchmark’s original test partition is subdivided for calibration, explicitly identify the reserved subset and exclude it from final reporting. More complex tuning can use nested validation or a separate selection split. A benchmark’s original partition name is less informative than how each example was actually used.

Resource matching also requires more than equal annotation totals. Preference-distribution learning in [38] compares 50,000 annotation calls allocated as 5,000 inputs with ten labels each for SFT and 50,000 inputs with one label each for RL. This is an informative sparse-versus-dense supervision comparison, with tenfold different input coverage. It

does not isolate the optimizer under identical inputs. Likewise, a transferred calibrator may use no target labels while still requiring substantial labeled source data [97].

Decision evaluation should include two operating conditions. First, apply a probability threshold determined by declared costs, testing the reported numerical meaning directly. Second, select a threshold on independent validation data to test the achievable usefulness of the score for that deployment. Present the resulting coverage and error costs on test data. A ranking method may perform well in the second condition despite lacking an interpretable probability scale. Attaining a target accuracy in one small accepted subset is empirical evidence, rather than a distribution-free certificate [25] (Table 3).

The elicitation protocol belongs in this contract. The ownership-framing study holds candidate answers fixed while changing whether they appear as assistant output or user-provided content; the resulting confidence changes identify a contextual confound [7]. It also scores all answer options, unlike top-answer-only evaluations. Broader uncertainty evaluations compare token likelihood, numerical confidence, and linguistic uncertainty extracted by another model; these channels differ in information access, parsing failures, and inference cost [98]. A reproducible comparison therefore retains the answer text, role/template, selected-versus-all-candidate population, parse-failure outcomes, and any auxiliary judge calls.

Refusal further changes the evaluated population. Reasoning-model comparisons that calculate calibration only on attempted answers should also report all-query coverage and a shared-attempted comparison [5]. The Hallucination Tax study evaluates refusal on explicitly unanswerable questions [6], which supplies decision evidence distinct from probability calibration. The RLVR measurement position paper brings these requirements together: attempted-answer counts, shared-set stated confidence, and matched inference budgets can change how post-training comparisons are interpreted [99]. Its heterogeneous checkpoint and budget comparisons motivate controlled measurements rather than a universal direction of calibration change.

6.3. What within-study comparisons establish

The following examples illustrate different claims; their numbers should not be combined into a cross-study leaderboard.

Table 5: Within-study evidence of distinct improvements; no cross-study ranking is implied.

Within-study comparison	Observed result	Supported interpretation
Qwen3-4B training comparison [4] (Tables 1–2)	Mean ECE is 0.163 initially, 0.135 after GRPO, 0.117 after DPO, and 0.034 after SFT	GRPO/DPO outperform initialization on this metric but remain worse than the supervised comparator
EpiCaR transfer, Qwen3-8B on GSM8K [100] (Table 3)	AUROC rises 0.654→0.722 while raw ECE rises 0.215→0.364	Better error ordering does not establish better probability scaling
RLSR versus RLCR, HotPotQA-trained Qwen2.5-7B in-domain aggregate [25] (Table 1)	AURC improves 0.51→0.44 and accuracy at 10% coverage 53.5%→66.0%, while ECE rises 0.05→0.24	A selective-ranking objective can improve its intended behavior without improving calibration
Gemma-2-9B preference evaluation [38] (Table 1)	RL-Brier has lower soft-target MSE than temperature scaling, 0.0764 versus 0.0839, but higher reported ECE, 0.0879 versus 0.0827	Soft preference proportions and tie-excluded majority labels are different evaluation targets
RLVP-Anneal versus XGBoost on six single tasks [22] (Table 1)	AUC improves on 6/6 tasks; ECE improves on 2/6. Unweighted mean ECE is 0.1005 versus 0.0743, calculated from the reported rows	A useful ranking gain is not uniform calibration dominance

The first row also illustrates why a paper’s broad verbal diagnosis should be read alongside its comparator. Although the study describes RL/DPO-induced miscalibration, its reported mean ECE improves relative to initialization and remains worse than SFT [4]. The conclusion supported by that table is comparator-specific.

These distinctions should be credited accurately. EpiCaR explicitly discusses discrimination versus absolute calibration, and RLSR is designed around selective ranking. Their results motivate evaluation against each method’s intended objective. The preference example goes further: its metric disagreement changes the target itself, rather than merely presenting two summaries of the same probability error.

Version-aware reading can also strengthen an evaluation. The ACL version of decision-token calibration [31] adds AUROC and calibration-weight/KL sensitivity results (Tables 7–9). For Qwen3-1.7B, increasing the calibration weight from 0.001 to 0.005 reduces CSQA ECE from 15.97 to 4.26, while OBQA ECE increases from 5.27 to 9.60; ECE is reported on a 0–100 scale. The result supports sensitivity to weighting and transfer conditions. The same publication still selects its best checkpoint on the test set (Appendix A), so the added analyses do not establish independent test selection.

EpiCaR presents a different issue. Its appendix reserves 500 original test examples for temperature fitting and reports on the remainder, which is a legitimate holdout design as described. However, one passage specifies minimizing ECE and another NLL [100]. This documentation conflict should be resolved or reported; it should not be conflated with checkpoint selection on the final test results.

RLVP also shows why the population and metric must travel with a headline result. Its “up to 55%” teacher-surpass ratio concerns per-task AUC on the 169 jointly trained tasks’ test splits [22] (Figs. 2–3). The eight unseen tasks in that paper’s Table 2 instead report AUC alone: zero-shot and 32-shot RLVP beat the 32-example XGBoost comparator on four and five tasks, respectively. Those results support transfer of discrimination, while calibration transfer to that population requires corresponding probability-quality measurements. The six-task ECE values in Table 1 also do not support the paper’s broad prose claim of best average ECE. These qualifications retain the positive results without extending their evidential scope.

6.4. Metrics, implementation, and uncertainty

Metric names are insufficient identifiers. State the label, binning and weighting, report transformation, sample retention, and aggregation level. The audited papers and released evaluators include per-example absolute error labeled “ECE,” differing hard-versus-graded correctness targets, and paper/dependency differences in bin weighting [53], [72], [81]. The appendix material records these source-specific details and distinguishes inspected releases from historical experiment runs.

Finally, uncertainty estimates should respect the unit of dependence. Multiple trajectories from one question are not independent test questions; repeated annotations may also share an annotator or task. Use paired or clustered resampling accordingly, report the number of rare errors and accepted examples, and keep label uncertainty separate from training-seed variability. These practices turn a metric improvement into an interpretable result: a better reporter on fixed outcomes, a better native policy, or a better decision under specified costs.

7. Research agenda: open questions about mechanisms and decisions

The synthesis identifies several explanations for improved confidence metrics: better reporting, changed answer policies, different effective objectives, and better operating-point selection. Existing author-reported studies provide evidence within their protocols; the analytic constructions establish possibilities under explicit assumptions. Neither establishes how frequently a mechanism explains other trained models. This survey reports no independently trained LLM results. The four questions below organize further inquiry around evidence that could distinguish explanations, building on the comparisons already available in the reviewed literature.

7.1. When does joint scoring change reporting or capability allocation?

Proper reporting incentives do not determine how a jointly trained policy distributes success across inputs. Section 3 separates mean capability, the distribution of conditional success probabilities, and reporting regret. Its coupled-policy construction shows that exact reward optimization can prefer lower mean accuracy even with truthful reports. With

truthful reports attainable, a feasible pointwise-dominating policy aligns the objectives when the calibrated value is nondecreasing. RLCR and reward-family comparisons provide existing evidence about joint training [19], [20]; identifying which decomposition term explains a particular outcome requires more than aggregate ECE and accuracy.

The informative comparisons distinguish changes in objective preference from restricted reporting capacity and finite optimization. Settings with known success probabilities can vary reward curvature and accuracy weight across coupled and independently adjustable policy families. Attainable truthful reports isolate allocation; a pointwise-dominating policy supplies a negative control under the conditions above. The bounded-regret result constrains exact optima, not neural convergence. For LLM evidence, common input weights, decoders and verifiers, together with uncertainty estimates for conditional success profiles, make this attribution more credible. Compatible crossed reporters on fixed answer pools can identify reporting changes separately from native-policy gains, while selective utility establishes whether a changed capability profile benefits the intended decision.

7.2. Which effects belong to the score, and which to the update?

A population scoring objective, its finite-sample estimate and its implemented gradient need not have the same properties. Conventional empirical CRPS uses the pairwise factor $1/(2K^2)$. For $K \geq 2$, replacing it with $1/[2K(K-1)]$ yields an unbiased estimate of population CRPS under iid ensemble draws, finite first moments, and independence from the realized outcome conditional on the input [16]. The Gaussian construction establishes that the conventional score can favor insufficient spread before neural optimization enters. It does not establish that this mechanism explains DAR’s reported results: DAR adds leave-one-out credit, normalization and valid-sample handling [27].

Author-reported normalization ablations [30] and REAL’s direct-gradient ablation [26] already support examining update choices, but their conclusions remain specific to the evaluated targets and configurations. Further attribution would compare raw scores, rollout credits, transformed advantages and gradients on common samples. Varying K with explicit sampling budgets distinguishes estimator effects from additional computation; separating centering from standardization distinguishes baseline subtraction from random rescaling. Dependent generation and parse filtering require additional analysis beyond the independent-draw reference. For parameter-dependent rewards, the exact gradient with both score-function and direct terms, a stopped-gradient reward and a stabilized estimator are distinct comparisons. Fairness of the scalar score alone does not certify any transformed gradient, and differences between published formulas and inspected launchers cannot identify historical run settings.

7.3. When do reusable probability magnitudes add decision value?

Probability reporting, selective ranking and direct action learning serve overlapping deployment needs. The four-context construction fixes answers and outcomes and shows how a strictly increasing transformation preserves ordering and attainable selection frontiers while changing utility at a transported numerical threshold. Reselecting the threshold can recover the accepted set without recalibrating the probabilities. Existing RLSR and TruthRL comparisons supply evidence for selective ranking and action learning at their respective operating conditions [25], [28]; they do not establish a universal preference among these strategies.

The unresolved comparison concerns transfer across costs, contexts and available labels. On common answers, cost-derived thresholds assess numerical probability meaning, whereas thresholds selected using independent validation labels assess attainable deployment performance. Native-policy evaluation then retains any benefit from changing answers. Transfer to withheld costs without evaluation-label reuse could establish when calibrated, informative reports reduce additional supervision, when a rank score suffices after threshold fitting, or when direct action learning remains preferable at a fixed utility. Contextual costs and abstention-induced missing labels must enter that comparison explicitly, because they change the information needed for a valid decision [13].

7.4. Which uncertainty estimate supports an adaptive decision?

Confidence in a completed answer and the expected benefit of another action concern different events and information sets. Section 5 distinguishes response confidence from decoder-specific capability. The capability study provides evidence for matching the predictor to a repeated-success target [36]; its reported pass@64 simulation does not by itself validate a deployed selector. Even under independent trials, finding at least one correct candidate differs from recognizing and selecting it.

An informative adaptive-computation comparison would retain pre-action states, available information, chosen actions, observed costs, candidate outcomes and the selection rule. Comparing heuristics, supervised value predictors, capability estimates and RL controllers under the same information and budget could distinguish better prediction from additional information or computation. A held-out decoder, policy or budget regime tests the scope of transfer. Where information sets are nested and the outcome remains fixed, the conditional-expectation identity in Appendix A provides a coherence check. Where retrieval, routing or continued generation changes the outcome distribution, the resulting policy and its cost must be evaluated directly. The open question is which probability target remains useful at the moment each intervention is chosen.

8. Conclusion

Calibration-aware RL is best understood through the relationship between a probability target, its scoring incentive, the implemented update, and the decision made from its output. Proper scores provide a principled reporting objective, but joint answer training also values the distribution of conditional success probabilities. Finite ensembles and group transformations introduce further distinctions between a population score and practical learning. Downstream, useful ranking and reliable probability magnitudes solve overlapping but different problems.

These distinctions lead to concrete choices. Post-hoc fitting is a necessary comparator when predictions can remain fixed. Direct supervision is a necessary comparator when the desired report or statistic has an available differentiable loss. RL becomes a distinct intervention when generation, information acquisition, or trajectory decisions must change under feedback. Evaluation should then isolate reporting, capability, estimator behavior, and operating-point selection under a common protocol. The central opportunity is to connect improvements at those individual stages to evidence of reliable decisions, with enough detail to explain both when the connection holds and which comparison is needed when it does not.

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References

- [1] Diogo Almeida / TypeSafe, “Introducing System One Models & Jev.” 2026. Available: <https://typesafe.ai/bl/og/introducing-system-one-models-and-jev>
- [2] OpenAI *et al.*, “GPT-4 Technical Report.” 2023. Available: <https://arxiv.org/abs/2303.08774>
- [3] D. Yoon *et al.*, “Reasoning Models Better Express Their Confidence,” in *Advances in Neural Information Processing Systems*, 2025, pp. 103869–103896. doi: [10.52202/085713-3469](https://doi.org/10.52202/085713-3469).
- [4] X. Xie, X. Liu, and B. Yao, “Know When You’re Wrong: Aligning Confidence with Correctness for LLM Error Detection.” 2026. Available: <https://arxiv.org/abs/2603.06604>
- [5] Q. Zeng, W. Xuan, L. Cui, and R. Voigt, “Thinking Out Loud: Do Reasoning Models Know When They’re Right?” in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 2025, pp. 1394–1407. doi: [10.18653/v1/2025.emnlp-main.73](https://doi.org/10.18653/v1/2025.emnlp-main.73).
- [6] L. Song, T. Shi, and J. Zhao, “The Hallucination Tax of Reinforcement Finetuning,” in *Findings of the Association for Computational Linguistics: EMNLP 2025*, 2025, pp. 2105–2120. doi: [10.18653/v1/2025.findings-emnlp.112](https://doi.org/10.18653/v1/2025.findings-emnlp.112).
- [7] M. Sanz-Guerrero, M. Mager, and K. von der Wense, “Large Language Models Are Overconfident in Their Own Responses,” in *Findings of the Association for Computational Linguistics: ACL 2026*, 2026, pp. 31406–31418. doi: [10.18653/v1/2026.findings-acl.1570](https://doi.org/10.18653/v1/2026.findings-acl.1570).
- [8] A. Wang, W. Ma, and Y. Liu, “Let the Model Distribute Its Doubt: Confidence Estimation through Verbalized Probability Distribution.” 2025. Available: <https://arxiv.org/abs/2511.14275>

- [9] J. Geng, F. Cai, Y. Wang, H. Koepl, P. Nakov, and I. Gurevych, “A Survey of Confidence Estimation and Calibration in Large Language Models,” in *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, 2024, pp. 6577–6595. doi: [10.18653/v1/2024.naacl-long.366](https://doi.org/10.18653/v1/2024.naacl-long.366).
- [10] P.-C. Pan, Y. Liang, and S. Lin, “Reward Modeling for Reinforcement Learning-Based LLM Reasoning: Design, Challenges, and Evaluation,” *Transactions on Machine Learning Research*, 2026, Available: <https://openreview.net/forum?id=TDfrN1TbGH>
- [11] J. Zhang *et al.*, “From Passive Metric to Active Signal: The Evolving Role of Uncertainty Quantification in Large Language Models,” in *Findings of the Association for Computational Linguistics: ACL 2026*, 2026, pp. 41525–41544. doi: [10.18653/v1/2026.findings-acl.2064](https://doi.org/10.18653/v1/2026.findings-acl.2064).
- [12] O. Shorinwa, Z. Mei, J. Lidard, A. Z. Ren, and A. Majumdar, “A Survey on Uncertainty Quantification of Large Language Models: Taxonomy, Open Research Challenges, and Future Directions.” 2024. Available: <https://arxiv.org/abs/2412.05563>
- [13] B. Wen *et al.*, “Know Your Limits: A Survey of Abstention in Large Language Models,” *Transactions of the Association for Computational Linguistics*, vol. 13, pp. 529–556, 2025, doi: [10.1162/tacl_a_00754](https://doi.org/10.1162/tacl_a_00754).
- [14] Y. Li, T. Zhou, X. Shen, Y. Miao, R. Padman, and R. Krishnan, “Confidence as Control: A Survey of Confidence Utilization in Large Language Models.” 2026. Available: https://yubol-bobo.github.io/assets/pdf/Conf_Survey.pdf
- [15] T. Gneiting and A. E. Raftery, “Strictly Proper Scoring Rules, Prediction, and Estimation,” *Journal of the American Statistical Association*, vol. 102, no. 477, pp. 359–378, 2007, doi: [10.1198/016214506000001437](https://doi.org/10.1198/016214506000001437).
- [16] C. A. T. Ferro, “Fair scores for ensemble forecasts,” *Quarterly Journal of the Royal Meteorological Society*, vol. 140, no. 683, pp. 1917–1923, 2014, doi: [10.1002/qj.2270](https://doi.org/10.1002/qj.2270).
- [17] B. Turtel, P. Wilczewski, K. Skotheim, V. A. Satopää, and P. E. Tetlock, “How Proper Scoring Rules Shape LLM Forecasting.” 2026. Available: <https://arxiv.org/abs/2608.28482>
- [18] D. Bani-Harouni *et al.*, “Rewarding Doubt: A Reinforcement Learning Approach to Calibrated Confidence Expression of Large Language Models,” in *International Conference on Learning Representations*, 2026, pp. 117927–117943. Available: https://proceedings.iclr.cc/paper_files/paper/2026/hash/bf8065446507b0d3842838564ac4f1f3-Abstract-Conference.html
- [19] M. Damani *et al.*, “Beyond Binary Rewards: Training LMs to Reason About Their Uncertainty,” in *International Conference on Learning Representations*, 2026, pp. 59456–59493. Available: https://proceedings.iclr.cc/paper_files/paper/2026/hash/615675cc6e94ddb1a783904fb178b5f6-Abstract-Conference.html
- [20] C. H. Tan, Z. Lin, M. Motani, and W. S. Lee, “On the effectiveness of reward functions in reinforcement learning for confidence calibration of large language models.” 2026. Available: <https://arxiv.org/abs/2607.04332>
- [21] J. Leng, C. Huang, B. Zhu, and J. Huang, “Taming Overconfidence in LLMs: Reward Calibration in RLHF,” in *International Conference on Learning Representations*, 2025, pp. 16484–16517. Available: https://proceedings.iclr.cc/paper_files/paper/2025/hash/29fb6e1456b3d8b57ede5c45aa2c6537-Abstract-Conference.html
- [22] L. Li *et al.*, “Reinforcement Learning with Verbalized Probabilities for LLM Classification,” in *Proceedings of the ACM Web Conference 2026*, 2026, pp. 7401–7411. doi: [10.1145/3774904.3792540](https://doi.org/10.1145/3774904.3792540).
- [23] G. K.-M. Liu, A. Caciularu, G. Yona, I. Szepkter, and A. Cohan, “Reinforcement Learning with Metacognitive Feedback Elicits Faithful Uncertainty Expression in LLMs.” 2026. Available: <https://arxiv.org/abs/2606.32032>
- [24] Z. Wang, X. Lou, M. Wu, Z. Wen, and J. Zhang, “Calibration-Aware Policy Optimization for Reasoning LLMs,” in *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2026, pp. 18375–18390. doi: [10.18653/v1/2026.acl-long.836](https://doi.org/10.18653/v1/2026.acl-long.836).
- [25] G. Luo, Y. Wu, S. Zhang, A. Deshwal, and J. Sun, “Aligning Language Models with Selective Prediction.” 2026. Available: <https://arxiv.org/abs/2607.03528>
- [26] Y. Zhang, T. Chen, M. Zhou, O. Leong, Y. N. Wu, and M. Lukasik, “REAL: Regression-Aware Reinforcement Learning for LLM-as-a-Judge.” 2026. Available: <https://arxiv.org/abs/2603.17145>
- [27] J. Park *et al.*, “Distribution-Aware Reward: Reinforcement Learning over Predictive Distributions for LLM Regression.” 2026. Available: <https://arxiv.org/abs/2605.20740>
- [28] Z. Wei *et al.*, “TruthRL: Incentivizing Truthful LLMs via Reinforcement Learning,” in *Forty-Third International Conference on Machine Learning*, 2026. Available: <https://icml.cc/virtual/2026/poster/63785>

- [29] J. Wu, J. Liu, Z. Zeng, T. Zhan, T. Cai, and W. Huang, “Mitigating LLM Hallucination via Behaviorally Calibrated Reinforcement Learning.” 2025. Available: <https://arxiv.org/abs/2512.19920>. Presented as a poster at the non-archival ICLR 2026 Workshop "Agentic AI in the Wild: From Hallucinations to Reliable Autonomy", 27 April 2026. This is not an ICLR main-conference proceedings publication.
- [30] M. Bereket and J. Leskovec, “Uncalibrated Reasoning: GRPO Induces Overconfidence for Stochastic Outcomes.” 2025. Available: <https://arxiv.org/abs/2508.11800>
- [31] D. N. Yaldiz, E. Spiliopoulou, Z. Qi, S. Varia, S. Doss, and N. Pappas, “Balancing Classification and Calibration Performance in Decision-Making LLMs via Calibration Aware Reinforcement Learning,” in *Findings of the Association for Computational Linguistics: ACL 2026*, 2026, pp. 12537–12553. doi: [10.18653/v1/2026.findings-acl.610](https://doi.org/10.18653/v1/2026.findings-acl.610).
- [32] Y. Li, M. Xiong, J. Wu, and B. Hooi, “ConfTuner: Training Large Language Models to Express Their Confidence Verbally,” in *Advances in Neural Information Processing Systems*, 2025, pp. 53484–53513. doi: [10.52202/085713-1783](https://doi.org/10.52202/085713-1783).
- [33] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On Calibration of Modern Neural Networks,” in *Proceedings of the 34th International Conference on Machine Learning*, 2017, pp. 1321–1330. Available: <https://proceedings.mlr.press/v70/guo17a.html>
- [34] C. Li *et al.*, “Confidence Before Answering: A Paradigm Shift for Efficient LLM Uncertainty Estimation.” 2026. Available: <https://arxiv.org/abs/2603.05881>
- [35] C. Finlay, J. Kurien, S. Dash, M. Fadaee, and B. Ermis, “CALIBER: Calibrating Confidence Before and After Reasoning in Language Models.” 2026. Available: <https://arxiv.org/abs/2606.24281>
- [36] S.-H. Yang, C.-K. Wu, C.-Y. Lin, Y.-N. Chen, H. Lee, and S.-H. Sun, “On Calibration of Large Language Models: From Response To Capability.” 2026. Available: <https://arxiv.org/abs/2602.13540>
- [37] X. Yang, Y. Yuan, S. Lei, and X. Wang, “Scaling with Confidence: Calibrating Confidence of LLMs for Adaptive Test Time Scaling.” 2026. Available: <https://arxiv.org/abs/2607.01612>
- [38] Z. Li *et al.*, “Judging with Confidence: Calibrating Autoraters to Preference Distributions.” 2025. Available: <https://arxiv.org/abs/2510.00263>
- [39] S. Kadavath *et al.*, “Language Models (Mostly) Know What They Know.” 2022. Available: <https://arxiv.org/abs/2207.05221>
- [40] S. Lin, J. Hilton, and O. Evans, “Teaching Models to Express Their Uncertainty in Words,” *Transactions on Machine Learning Research*, 2022, Available: <https://openreview.net/forum?id=8s8K2UZGTZ>
- [41] K. Tian *et al.*, “Just Ask for Calibration: Strategies for Eliciting Calibrated Confidence Scores from Language Models Fine-Tuned with Human Feedback,” in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023, pp. 5433–5442. doi: [10.18653/v1/2023.emnlp-main.330](https://doi.org/10.18653/v1/2023.emnlp-main.330).
- [42] J. Bröcker, “Reliability, Sufficiency, and the Decomposition of Proper Scores,” *Quarterly Journal of the Royal Meteorological Society*, vol. 135, no. 643, pp. 1512–1519, 2009, doi: [10.1002/qj.456](https://doi.org/10.1002/qj.456).
- [43] T. Gneiting, F. Balabdaoui, and A. E. Raftery, “Probabilistic forecasts, calibration and sharpness,” *Journal of the Royal Statistical Society: Series B*, vol. 69, no. 2, pp. 243–268, 2007, doi: [10.1111/j.1467-9868.2007.00587.x](https://doi.org/10.1111/j.1467-9868.2007.00587.x).
- [44] Y. Geifman and R. El-Yaniv, “Selective Classification for Deep Neural Networks,” in *Advances in Neural Information Processing Systems*, 2017. Available: https://papers.nips.cc/paper_files/paper/2017/hash/4a8423d5e91fda00bb7e46540e2b0cf1-Abstract.html
- [45] Y. Geifman, G. Uziel, and R. El-Yaniv, “Bias-Reduced Uncertainty Estimation for Deep Neural Classifiers,” in *International Conference on Learning Representations*, 2019. Available: <https://openreview.net/forum?id=SJfb5jCqKm>
- [46] A. N. Angelopoulos and S. Bates, “A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification.” 2021. Available: <https://arxiv.org/abs/2107.07511>
- [47] A. T. Kalai and S. S. Vempala, “Calibrated Language Models Must Hallucinate,” in *Proceedings of the 56th Annual ACM Symposium on Theory of Computing*, 2024, pp. 160–171. doi: [10.1145/3618260.3649777](https://doi.org/10.1145/3618260.3649777).
- [48] N. Chandak, S. Goel, A. Prabhu, M. Hardt, and J. Geiping, “Scaling Open-Ended Reasoning To Predict the Future.” 2025. Available: <https://arxiv.org/abs/2512.25070>
- [49] B. Turtel, P. Wilczewski, D. Franklin, and K. Skothiem, “Future-as-Label: Scalable Supervision from Real-World Outcomes.” 2026. Available: <https://arxiv.org/abs/2601.06336>

- [50] B. Turtel, D. Franklin, K. Skotheim, L. Hewitt, and P. Schoenegger, “Outcome-based Reinforcement Learning to Predict the Future,” *Transactions on Machine Learning Research*, 2025, Available: <https://openreview.net/forum?id=bbhdeL8EUX>
- [51] I. Puri, M. Damani, I. Shenfeld, M. Ghassemi, J. Andreas, and Y. Kim, “Reaching Beyond the Mode: RL for Distributional Reasoning in Language Models.” 2026. Available: <https://arxiv.org/abs/2603.24844>
- [52] Z. Ma *et al.*, “Decoupling Reasoning and Confidence: Resurrecting Calibration in Reinforcement Learning from Verifiable Rewards,” in *Forty-Third International Conference on Machine Learning*, 2026. Available: <https://icml.cc/virtual/2026/poster/66459>
- [53] T. Xu *et al.*, “SaySelf: Teaching LLMs to Express Confidence with Self-Reflective Rationales,” in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 2024, pp. 5985–5998. doi: 10.18653/v1/2024.emnlp-main.343.
- [54] F. Yu, R. Niu, D. Wu, Y. Ma, and R. Yu, “Calibrating LLMs with Semantic-level Reward.” 2026. Available: <https://arxiv.org/abs/2605.15588>
- [55] X. Liu and L. Wang, “Think Through Uncertainty: Improving Long-Form Generation Factuality via Reasoning Calibration.” 2026. Available: <https://arxiv.org/abs/2604.12046>
- [56] W. Xuan, Q. Zeng, H. Qi, Y. Xiao, J. Wang, and N. Yokoya, “The Confidence Dichotomy: Analyzing and Mitigating Miscalibration in Tool-Use Agents.” 2026. Available: <https://arxiv.org/abs/2601.07264>
- [57] S. Tao *et al.*, “When to Trust LLMs: Aligning Confidence with Response Quality,” in *Findings of the Association for Computational Linguistics: ACL 2024*, 2024, pp. 5984–5996. doi: 10.18653/v1/2024.findings-acl.357.
- [58] A. Jha, A. Mahajan, A. V. Aravindan, P. Saravanan, S. S. Policharla, and S. C. Gehlot, “Rewarding Intellectual Humility: Learning When Not to Answer in Large Language Models.” 2026. Available: <https://arxiv.org/abs/2601.20126>
- [59] L. Wang, C. Zuo, W. Jurayj, B. Van Durme, and A. Liu, “Process Supervision of Confidence Margin for Calibrated LLM Reasoning.” 2026. Available: <https://arxiv.org/abs/2604.23333>
- [60] P. Cui, B. Yang, and J. Zhu, “Ranking-Aware Calibration for Reliable Multimodal Reinforcement Learning.” 2026. Available: <https://arxiv.org/abs/2605.16999>
- [61] H. Liu, S. Wang, and H. Xu, “C²GSPG: Confidence-calibrated Group Sequence Policy Gradient towards Self-aware Reasoning.” 2025. Available: <https://arxiv.org/abs/2509.23129>
- [62] J. Huang, P. Lu, and Q. Zeng, “Calibrated Language Models and How to Find Them with Label Smoothing,” in *Proceedings of the 42nd International Conference on Machine Learning*, 2025, pp. 25593–25612. Available: <https://proceedings.mlr.press/v267/huang25w.html>
- [63] S. Kapoor *et al.*, “Large Language Models Must Be Taught to Know What They Don’t Know,” in *Advances in Neural Information Processing Systems*, 2024, pp. 85932–85972. doi: 10.52202/079017-2729.
- [64] J. Zhang, X. Peng, Q. Chen, Q. Ye, C. Xiong, and C.-S. Wu, “The Illusion of Certainty: Decoupling Capability and Calibration in On-Policy Distillation.” 2026. Available: <https://arxiv.org/abs/2604.16830>
- [65] X. Che, Y. Yuan, W. Zhao, and C. Yu, “Abstention as an Action Can Kill Both the Reward Gradient and the KL Anchor: Collapse Law and Repair for Error-Penalized Reinforcement Learning.” 2026. Available: <https://arxiv.org/abs/2608.00301>
- [66] X. Zeng *et al.*, “UCPO: Uncertainty-Aware Policy Optimization.” 2026. Available: <https://arxiv.org/abs/2601.22648>
- [67] R. Berdichevsky, S. Nahum-Gefen, and E. B. Zaken, “SALSA: Single-pass Autoregressive LLM Structured Classification.” 2025. Available: <https://arxiv.org/abs/2510.22691>
- [68] H. Liu, H. Huang, X. Gu, H. Wang, and Y. Wang, “On Calibration of LLM-based Guard Models for Reliable Content Moderation,” in *International Conference on Learning Representations*, 2025, pp. 67808–67829. Available: https://proceedings.iclr.cc/paper_files/paper/2025/hash/a99f732df9b668284b449da0214a3286-Abstract-Conference.html
- [69] V. Wang, M. J. Zhang, and E. Choi, “Improving LLM-as-a-Judge Inference with the Judgment Distribution,” in *Findings of the Association for Computational Linguistics: EMNLP 2025*, 2025, pp. 23173–23199. doi: 10.18653/v1/2025.findings-emnlp.1259.
- [70] S. Singh, A. Reddy, and M. Chopra, “Verifiable Rewards for Calibrated Probabilistic Forecasting.” 2026. Available: <https://arxiv.org/abs/2607.00164>

- [71] N. Band, X. Li, T. Ma, and T. Hashimoto, “Linguistic Calibration of Long-Form Generations,” in *Proceedings of the 41st International Conference on Machine Learning*, 2024, pp. 2732–2778. Available: <https://proceedings.mlr.press/v235/band24a.html>
- [72] Y. Xu *et al.*, “DualStake: Dual-Path Confidence Calibration in Deep Research Agents.” 2026. Available: <https://arxiv.org/abs/2609.00935>
- [73] Y. Zhu, “Closing the Reflection Gap: A Free Calibration Bonus for Agentic RL.” 2026. Available: <https://arxiv.org/abs/2606.14211>
- [74] A. Jenane, N. Walha, L. Kuhn, and F. Buettner, “From Entropy to Calibrated Uncertainty: Training Language Models to Reason About Uncertainty.” 2026. Available: <https://arxiv.org/abs/2603.06317>
- [75] P. S. Pandey and Z. Jin, “BinaryPPO: Efficient Policy Optimization for Binary Classification.” 2026. Available: <https://arxiv.org/abs/2602.02708>
- [76] M. Xiong *et al.*, “Can LLMs Express Their Uncertainty? An Empirical Evaluation of Confidence Elicitation in LLMs,” in *International Conference on Learning Representations*, 2024. Available: <https://openreview.net/forum?id=gjeQKFxFpZ>
- [77] J. Xiao *et al.*, “Restoring Calibration for Aligned Large Language Models: A Calibration-Aware Fine-Tuning Approach,” in *Proceedings of the 42nd International Conference on Machine Learning*, 2025, pp. 68364–68390. Available: <https://proceedings.mlr.press/v267/xiao25b.html>
- [78] N. Parikh, A. Sai, P. Shivaswamy, K. Panchal, and A. Lan, “CATTO: Balancing Preferences and Confidence in Language Models.” 2026. Available: <https://arxiv.org/abs/2601.23096>
- [79] H. Xiao, V. Patil, H. Lee, E. Stengel-Eskin, and M. Bansal, “Generalized Correctness Models: Learning Calibrated and Cross-Model Correctness Predictors from Historical Patterns,” in *Forty-Third International Conference on Machine Learning*, 2026. Available: <https://icml.cc/virtual/2026/poster/62488>
- [80] B. Luo, C. Wang, H. Wei, S. Li, and X. Du, “Unlocking the Pre-Trained Model as a Dual-Alignment Calibrator for Post-Trained LLMs.” 2026. Available: <https://arxiv.org/abs/2601.04277>
- [81] E. Stengel-Eskin, P. Hase, and M. Bansal, “LACIE: Listener-Aware Finetuning for Calibration in Large Language Models,” in *Advances in Neural Information Processing Systems*, 2024, pp. 43080–43106. doi: 10.52202/079017-1364.
- [82] X. Zhao, Z. Kang, A. Feng, S. Levine, and D. Song, “Learning to Reason without External Rewards,” in *International Conference on Learning Representations*, 2026, pp. 2548–2581. Available: https://proceedings.iclr.cc/paper_files/paper/2026/hash/052e2404709e8df9dc38fbc7bfec5b80-Abstract-Conference.html
- [83] R. Jin *et al.*, “Revisiting Entropy in Reinforcement Learning for Large Reasoning Models,” in *Findings of the Association for Computational Linguistics: ACL 2026*, 2026, pp. 25300–25322. doi: 10.18653/v1/2026.findings-acl.1266.
- [84] Q. Yu *et al.*, “DAPO: An Open-Source LLM Reinforcement Learning System at Scale,” in *Advances in Neural Information Processing Systems*, 2025, pp. 113222–113244. doi: 10.52202/085713-3775.
- [85] Z. Liu *et al.*, “Understanding R1-Zero-Like Training: A Critical Perspective,” in *Conference on Language Modeling*, 2025. Available: <https://openreview.net/forum?id=jLpC1zavzn>
- [86] C. Wang, G. Szarvas, G. Balazs, P. Danchenko, and P. Ernst, “Calibrating Verbalized Probabilities for Large Language Models.” 2024. Available: <https://arxiv.org/abs/2410.06707>
- [87] A. Chagini *et al.*, “Reasoning’s Razor: Reasoning Improves Accuracy but Can Hurt Recall at Critical Operating Points in Safety and Hallucination Detection.” 2025. Available: <https://arxiv.org/abs/2510.21049>
- [88] A. T. Kalai, O. Nachum, S. S. Vempala, and E. Zhang, “Why Language Models Hallucinate.” 2025. Available: <https://arxiv.org/abs/2509.04664>
- [89] I. Stepanov, M. Shtopko, D. Vodianytskyi, O. Lukashov, A. Yavorskyi, and M. Yaroshenko, “GLiClass: Generalist Lightweight Model for Sequence Classification Tasks.” 2025. Available: <https://arxiv.org/abs/2508.07662>
- [90] TypeSafe, “AI primer: RLCD and calibrated decisions.” 2026. Available: <https://docs.typesafe.ai/introduction/machine-learning-primer>
- [91] TypeSafe, “Confidence — TypeSafe.” 2026. Available: <https://docs.typesafe.ai/confidence>
- [92] harshatheg, “Qwen-2.5-1B-RLCD / Parallel Constrained Decoding for Apple Silicon.” 2026. Available: <https://huggingface.co/harshatheg/Qwen-2.5-1B-RLCD>

- [93] rorshopping, “Jev-style decisions on a laptop.” 2026. Available: <https://github.com/rorshopping/jev-on-a-laptop>
- [94] kikoncuo, “JEVfire.” 2026. Available: <https://github.com/kikoncuo/jevfire>
- [95] AlexWortega, “openjev — Qwen3.5 trained as jev model.” 2026. Available: <https://huggingface.co/AlexWortega/openjev>
- [96] K. Park, J. Wang, T. Berg-Kirkpatrick, N. Polikarpova, and L. D’Antoni, “Grammar-Aligned Decoding,” in *Advances in Neural Information Processing Systems*, 2024. Available: https://proceedings.neurips.cc/paper_files/paper/2024/file/2bdc2267c3d7d01523e2e17ac0a754f3-Paper-Conference.pdf
- [97] M. Shen, S. Das, K. Greenewald, P. Sattigeri, G. W. Wornell, and S. Ghosh, “Thermometer: Towards Universal Calibration for Large Language Models,” in *Proceedings of the 41st International Conference on Machine Learning*, 2024, pp. 44687–44711. Available: <https://proceedings.mlr.press/v235/shen24c.html>
- [98] L. Tao, Y.-F. Yeh, M. Dong, T. Huang, P. Torr, and C. Xu, “Revisiting Uncertainty Estimation and Calibration of Large Language Models.” 2025. Available: <https://arxiv.org/abs/2505.23854>
- [99] F. Wu *et al.*, “Position: The Hidden Costs and Measurement Gaps of Reinforcement Learning with Verifiable Rewards.” 2025. Available: <https://arxiv.org/abs/2509.21882>
- [100] J. Yeom, J. Sok, S. Park, J. Park, and T. Kim, “EpiCaR: Knowing What You Don’t Know Matters for Better Reasoning in LLMs.” 2026. Available: <https://arxiv.org/abs/2601.06786>
- [101] Scott Jeen; Matthew Aitchison; Mantic, “Training LLMs to Predict World Events (Guest Post with Mantic).” 2026. Available: <https://thinkingmachines.ai/news/training-llms-to-predict-world-events/>
- [102] S. Li *et al.*, “A Survey on the Honesty of Large Language Models,” *Transactions on Machine Learning Research*, 2025, Available: <https://openreview.net/forum?id=FJgtVfUxLQ>
- [103] Z. Xia, J. Xu, Y. Zhang, and H. Liu, “A Survey of Uncertainty Estimation Methods on Large Language Models,” in *Findings of the Association for Computational Linguistics: ACL 2025*, 2025, pp. 21381–21396. doi: [10.18653/v1/2025.findings-acl.1101](https://doi.org/10.18653/v1/2025.findings-acl.1101).
- [104] X. Liu, T. Chen, L. Da, C. Chen, Z. Lin, and H. Wei, “Uncertainty Quantification and Confidence Calibration in Large Language Models: A Survey,” in *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2*, 2025. doi: [10.1145/3711896.3736569](https://doi.org/10.1145/3711896.3736569).
- [105] L. Xie *et al.*, “A Survey of Calibration Process for Black-Box LLMs.” 2024. Available: <https://arxiv.org/abs/2412.12767>
- [106] S. Kang, Y. F. Bakman, D. N. Yaldiz, B. Buyukates, and S. Avestimehr, “Uncertainty Quantification for Hallucination Detection in Large Language Models: Foundations, Methodology, and Future Directions.” 2025. Available: <https://arxiv.org/abs/2510.12040>

Appendix A. Mathematical analysis and exact examples

These derivations organize the reviewed methods using classical conditional expectation and proper scoring rules [15], [16]. They are not claims of new general statistical theorems. Constructed distributions are distinguished from empirical LLM results throughout. Reward signs, admissible reports, information sets, and the distribution over which expectations are taken are part of each statement.

A.1. Reporting error and available information

Let $Z \in \{0, 1\}$, let $\mathcal{F}$ be a sigma-field describing the reporter’s information, and write $p_{\mathcal{F}} = \mathbb{E}[Z \mid \mathcal{F}]$. For any $\mathcal{F}$ -measurable report $C \in [0, 1]$,

$$\mathbb{E}(Z - C)^2 = \mathbb{E}[p_{\mathcal{F}}(1 - p_{\mathcal{F}})] + \mathbb{E}(C - p_{\mathcal{F}})^2. \quad (\text{A.1})$$

Expand $Z - C = (Z - p_{\mathcal{F}}) + (p_{\mathcal{F}} - C)$. The cross term vanishes conditional on $\mathcal{F}$. The first term measures uncertainty remaining under that information; the second is excess Brier risk relative to its conditional probability. It is not an empirical ECE estimator.

For nested information $\mathcal{F} \subseteq \mathcal{H}$, with the same outcome and joint distribution, the tower property gives

$$\mathbb{E}[p_{\mathcal{H}} \mid \mathcal{F}] = p_{\mathcal{F}}, \quad \mathbb{E}(Z - p_{\mathcal{F}})^2 - \mathbb{E}(Z - p_{\mathcal{H}})^2 = \mathbb{E}(p_{\mathcal{H}} - p_{\mathcal{F}})^2. \quad (\text{A.2})$$

This is a coherence condition for ideal reports before and after receiving information. A pre-generation capability estimate and a post-generation report can be compared this way only if they refer to the same realized correctness outcome under the same generation policy. A policy change, revised verifier, or different population changes the comparison.

Score calibration, $\mathbb{E}[Z \mid C] = C$, is weaker than $C = p_{\mathcal{F}}$. A constant base-rate report can be calibrated while ignoring predictive information. Two calibrated scalar scores therefore do not automatically satisfy a tower relationship. Likewise, uncertainty arising from computational limitations should not be conflated with residual randomness under complete semantic conditioning. If the conditioning includes a complete prompt, complete answer, and deterministic correctness rule, the corresponding conditional label probability is 0 or 1. Intermediate probabilities require a specified information restriction, latent or stochastic labels, or an action denoting a stochastic answer procedure.

A.2. Bayes reward and the joint-objective decomposition

For a reward-oriented proper binary score $S(c, z)$, define

$$L(c, p) = pS(c, 1) + (1 - p)S(c, 0), \quad G(p) = L(p, p), \quad D(p, c) = G(p) - L(c, p) \geq 0.$$

Strict propriety makes the zero-regret report unique within the stated admissible domain. Under the regular Savage representation, this is the Bregman regret $D_G(p, c) = G(p) - G(c) - g(c)(p - c)$, where $g(c)$ is an appropriate subgradient [15]. The decomposition below also follows directly from the definition $D = G - L$.

For one fixed policy π , let p_{π} be its conditional probability of success under the declared reporting information, and let c_{π} be a measurable report. Consider

$$R = \alpha Z + b + \beta S(c_{\pi}, Z), \quad \beta > 0,$$

with fixed finite weights and well-defined expectations. Writing $\mu_{\pi} = \mathbb{E}p_{\pi}$ gives

$$\begin{aligned} J(\pi, c_{\pi}) &= \alpha \mu_{\pi} + b + \beta G(\mu_{\pi}) \\ &\quad + \beta \{\mathbb{E}G(p_{\pi}) - G(\mu_{\pi})\} - \beta \mathbb{E}D_G(p_{\pi}, c_{\pi}). \end{aligned} \quad (\text{A.3})$$

The identity does not require truthful reports. It first conditions the expected score on the reporting information, then adds and subtracts $G(\mu_{\pi})$. Its three varying components are mean capability, the Jensen gap of the success-probability

profile, and reporting regret. The Jensen gap is generally not variance; variance alone does not order expectations of every convex function.

For correctness plus negative binary Brier, $\alpha = \beta = 1$, $b = 0$, $G(p) = p^2 - p$, and $D_G(p, c) = (p - c)^2$. Thus

$$J = \mathbb{E}[p_\pi^2] - \mathbb{E}(p_\pi - c_\pi)^2 = \mu_\pi^2 + \text{Var}(p_\pi) - \mathbb{E}(p_\pi - c_\pi)^2. \quad (\text{A.4})$$

This specializes the objective used in RLCR [19]. Its conditional truthful-report optimum is p^2 , increasing in p . Such pointwise preference does not settle comparisons among policies whose achievable success rates are coupled across inputs.

To isolate this issue, give two contexts equal weight and allow two candidate policies with independently attainable truthful reports:

Policy	Conditional success rates	Mean accuracy	Optimized joint reward	Utility at +1/−1/0
A	0.6, 0.6	0.600	0.36000	0.200
B	0.95, 0.2	0.575	0.47125	0.450

The joint reward prefers B despite its lower average accuracy. No reporting error or optimizer failure is present. Conversely, B has higher selective utility when correct, wrong, and rejected answers earn +1, −1, 0: accept only its first context, using threshold 0.5. The example demonstrates different objectives, rather than a universally undesirable outcome. It compares true conditional success across contexts, not random extreme confidence reports for one fixed event.

If reports are also constrained to $\mathcal{C}_\pi$, the ideal feasible value is

$$J_{\text{feasible}}^*(\pi) = \alpha\mu_\pi + b + \beta\mathbb{E}G(p_\pi) - \beta \inf_{c \in \mathcal{C}_\pi} \mathbb{E}D_G(p_\pi, c).$$

The final term can depend on the answer policy. It cannot be discarded as a common constant, and the resulting feasible value need not inherit a simple convex shape.

A.3. Curvature, capability constraints, and accuracy weights

Classical strict propriety implies strict convexity of the Bayes reward on a nondegenerate convex probability domain [15]. A short proof avoids differentiability. For $u \neq v$, $t \in (0, 1)$, and $m = tu + (1 - t)v$,

$$G(m) = L(m, m) = tL(m, u) + (1 - t)L(m, v) < tG(u) + (1 - t)G(v).$$

Both inequalities are strict because the report m differs from the truthful reports at u and v . Consequently $V(p) = \alpha p + b + \beta G(p)$ remains strictly convex for any fixed finite affine accuracy bonus and $\beta > 0$. A positive slope, as in pointwise best-effort conditions [20], can coexist with that curvature.

There is therefore no universal guarantee that expected reward preserves mean-accuracy rankings over every coupled feasible policy family. Choose interior $u < v$, let $m = (u + v)/2$, and define the strict Jensen gap $\Delta = [V(u) + V(v)]/2 - V(m) > 0$. Interior continuity permits a sufficiently small $\epsilon > 0$ with $V(m + \epsilon) < V(m) + \Delta$. Two equal-weight profiles $A = (m + \epsilon, m + \epsilon)$ and $B = (u, v)$ then satisfy

$$\text{Acc}(A) > \text{Acc}(B), \quad J^*(A) < J^*(B). \quad (\text{A.5})$$

The restricted family containing only A and B suffices. This is an elementary consequence of established score geometry, not an empirical explanation of any particular reported accuracy drop.

The quantifier matters: some families permit reversal; not every family does. If probabilities take only fixed endpoints 0 and 1, then $\mathbb{E}V(p) = V(0) + [V(1) - V(0)]\mathbb{E}p$, which preserves accuracy order when $V(1) > V(0)$. The same

affine argument applies to any fixed two-value support with positive secant slope. If each context can attain its best action independently, or one feasible policy dominates pointwise, a nondecreasing V permits joint optimization of reward and accuracy. Fixed external events also have no changing capability profile. Weak properness, nonconvex probability domains, report constraints, and richer non-affine mechanisms require their own analyses.

Increasing the accuracy weight remains useful. Suppose G is bounded, truthful reports are feasible, and exact optimizers exist over a common policy family. Let π_α maximize $\alpha \text{Acc}(\pi) + \beta \mathbb{E}G(p_\pi)$ for $\alpha > 0$, and let π_{acc} maximize accuracy. Optimality implies

$$0 \leq \text{Acc}(\pi_{\text{acc}}) - \text{Acc}(\pi_\alpha) \leq \frac{\beta \text{osc}(G)}{\alpha}, \quad \text{osc}(G) = \sup G - \inf G. \quad (\text{A.6})$$

Subtract the two objective values and bound the difference in expected G by its oscillation. For negative binary Brier on $[0, 1]$, $\text{osc}(G) = 1/4$. A fixed finite family with a positive accuracy gap can thus admit a sufficiently large weight selecting maximum accuracy, while no single finite weight provides exact ordering over all families with arbitrarily small gaps. This ideal bound does not guarantee finite neural optimization attains it.

A.4. Fixed-cost decision regret

Keep the answer/outcome distribution fixed, let $p = \mathbb{E}[Z \mid \mathcal{F}]$, and let c be $\mathcal{F}$ -measurable. Accepting a correct answer earns $b > 0$, accepting a wrong one loses $\ell > 0$, and rejection earns zero. The expected gain is $(b + \ell)(p - \tau)$, where $\tau = \ell/(b + \ell)$. Let $D_p = \mathbf{1}\{p \geq \tau\}$ and $D_c = \mathbf{1}\{c \geq \tau\}$. Their regret is

$$\mathcal{R}(D_c) = (b + \ell)\mathbb{E}[|p - \tau|\mathbf{1}\{D_c \neq D_p\}] \leq (b + \ell)\mathbb{E}|p - c| \leq (b + \ell)\sqrt{\mathbb{E}(p - c)^2}. \quad (\text{A.7})$$

On disagreement, τ lies between p and c , giving the first bound; Cauchy–Schwarz gives the second. Under the common-information condition, the squared term is excess Brier risk in (A.1). The result is not an ECE bound or a distribution-shift guarantee.

With only score calibration, $\mathbb{E}[Z \mid C] = C$, thresholding is optimal among policies observing that score under these fixed costs. It need not match a policy observing additional information or context-dependent costs. This distinction explains how an uninformative constant report can be calibrated without making good distinctions between cases.

A.5. Finite-ensemble CRPS

Let $X_1, \dots, X_K$ be independent draws from F , with finite first moments, and let y be a fixed outcome. Equivalently, condition on an input and require ensemble members to be independent of its realized outcome. For the empirical forecast distribution,

$$C_K = \frac{1}{K} \sum_i |X_i - y| - \frac{1}{2K^2} \sum_{i,j} |X_i - X_j|.$$

There are $K(K - 1)$ nonzero off-diagonal expectations and zero diagonal distances. Therefore

$$\mathbb{E}C_K = \text{CRPS}(F, y) + \frac{\mathbb{E}|X - X'|}{2K}. \quad (\text{A.8})$$

Replacing K^2 by $K(K - 1)$ in the pair term makes the score unbiased for population CRPS when $K > 1$ under these sampling conditions. Ferro calls the relevant property fairness for ensemble forecasts [16]; it concerns the generating distribution, rather than demographic fairness. A proper score for the empirical distribution and a fair score for its generating distribution are different requirements.

For a transparent analytic example, let truth be $Y \sim N(0, 1)$ and the predictive generator be $N(0, \sigma^2)$, with independent ensemble draws and outcome. Averaging conventional empirical CRPS gives

$$L_K(\sigma) = \sqrt{\frac{2}{\pi}} \sqrt{1 + \sigma^2} - \frac{1 - 1/K}{\sqrt{\pi}} \sigma, \quad \sigma_K^* = \frac{1 - 1/K}{\sqrt{2 - (1 - 1/K)^2}}. \quad (\text{A.9})$$

The optima at $K = 2, 4, 8$ are approximately 0.378, 0.626, and 0.788, matching the classical example in [16] (PDF p.5, Fig.2). The fair score has optimum 1. This is an analytic distributional illustration, not an LLM experiment. An implementation with dependent sampling, changing valid-sample counts, or additional reward transformations does not inherit an unbiased policy-gradient claim from (A.8). In particular, DAR’s leave-one-out credit and subsequent normalization require separate analysis [27].

A.6. Exact decision data and verification

The decision figure uses four equally probable contexts with success rates (0.1, 0.4, 0.6, 0.9). Truthful reports equal these rates; sharpened reports are (0.001, 0.1, 0.9, 0.999); the constant report is 0.5. Outcomes are integrated exactly rather than drawn as a finite sample.

Report	Population Brier	Population L1 reliability	AUROC	Coverage at 0.8	Utility at +1/ − 4/0
Truthful	0.1650000	0	0.825	0.25	0.125
Sharpened	0.2149005	0.1995	0.825	0.50	-0.125
Constant	0.2500000	0	0.500	0	0

The Brier decomposition [42] separates a common uncertainty term .25 from resolution and squared reliability error. Truthful and sharpened reports both have resolution .085 because their strict one-to-one mapping preserves exact-score information; their squared reliability errors are 0 and .0499005. The constant report has zero resolution and zero squared reliability error. These squared terms are distinct from the L1 values in the table.

The first two scores have the same ordering and tie structure, hence the same attainable threshold frontiers on this population. Reselecting a sharpened-score threshold strictly above 0.9 and no greater than 0.999 recovers the truthful acceptance set at 0.8. Equal AUROC alone, without the ordering condition, would not establish equal frontiers.

The released calculation scripts verify (A.1), (A.2), (A.7), and the Brier algebra on 100 exact rational tables; verify (A.3) on another 100 tables; enumerate all outcomes of discrete independent ensembles at $K = 2, \dots, 5$ to check (A.8); and check the analytic stationary points in (A.9). These checks validate the constructed examples and algebra. They do not reproduce any paper’s model results.

A.7. Exponentiating a score changes the population incentive

RLVP’s probability reward is $\exp\{ -[(1 - \lambda)D_{\text{KL}}(e_Y \| p) + \lambda D_{\text{KL}}(q \| p)] \}$, with λ annealed from 1 to 0 [22] (PDF p.3, Eq.2). The following is our analysis of that formula. Fix a teacher vector q , a realized label y , and $\lambda \in [0, 1]$, and define $m = (1 - \lambda)e_y + \lambda q$. On compatible supports,

$$\begin{aligned} (1 - \lambda)D_{\text{KL}}(e_y \| p) + \lambda D_{\text{KL}}(q \| p) &= D_{\text{KL}}(m \| p) + C, \\ C &= (1 - \lambda)D_{\text{KL}}(e_y \| m) + \lambda D_{\text{KL}}(q \| m) \geq 0. \end{aligned} \tag{A.10}$$

Thus the isolated probability component for this fixed sample is maximized at $p = m$, but its maximum is $\exp(-C)$, generally below one. Replacing the weighted KL sum by $D_{\text{KL}}(m \| p)$ alone drops a factor that can vary across outcomes and affect the composite reward. At the soft endpoint it matches the teacher; teacher fidelity does not establish fidelity to the true label distribution.

At the hard endpoint, $\exp[-D_{\text{KL}}(e_y \| p)] = p_y$. For an external label distribution r and a report issued before observing the label, the ungated expected component is $r^\top p$. With $r = (.7, .3)$, truthful reporting gives .58, whereas the strictly positive report (.999, .001) gives .6996. Both choose the same class, so an added argmax-correctness reward and a constant valid-format reward do not remove this difference. A proper log score instead rewards $\log p_y$, not p_y . More generally, pointwise monotonicity of a score transformation does not preserve its maximizer after averaging over outcomes.

This counterexample concerns the displayed component on an unrestricted report simplex. The paper’s teacher-correct gate changes where it applies, and the filtering implementation remains unresolved. It is not a prediction that finite

RLVP training converges to a concentrated report: the curriculum, parameter sharing, optimizer, and stopping rule all matter. Nor does the paper claim a propriety theorem for this reward; its explicit proper-scoring statement concerns the evaluation Brier loss. The accompanying P066 calculation check verifies (A.10), the endpoint reduction, and arithmetic transcribed from the result tables, without reproducing a model run.

Appendix B. Comparative method inventory

This inventory compares 49 learning, estimation, elicitation, and post-hoc source protocols, followed by three analyses with controlled learning experiments. Its 52 records distinguish protocols and analyses rather than counting distinct RL algorithms. Several papers contain variants; Table 2 instead compares learning families using selected examples. Theory, diagnostic studies, and deployment descriptions elsewhere in the survey remain evidence sources without being reclassified as prediction methods.

The three columns group labeled fields for the method and prediction object, supervision and update, and use with qualifications. The target and reporting interface remain distinct, as do feedback, objective, learning rule, and downstream use. Descriptions refer to each paper’s stated protocol unless a public-code qualification is explicit. Source locators below use **physical PDF pages** of the archived version; P007 uses its ACL version, and P089 its ICLR v2. Fixed-commit implementation checks and evaluation qualifications remain in Appendices C–D. The structured companion is `research/method_comparison_v0.4.csv`.

CE denotes cross-entropy; BCE, binary cross-entropy; KL, Kullback–Leibler divergence; SFT, supervised fine-tuning; RM, reward model; LOO, leave one out; and std, within-group reward standard deviation. PPO and GRPO name policy-optimization families rather than fully specifying losses or reductions. A numeric report, a class-token probability, and an action probability retain different statistical meanings even when they use the same optimizer.

B.1. Fixed outcomes and fixed-answer reporting

Method and prediction object	Supervision and update	Use and qualifications
Proper-score forecasting [17] RL / hybrid protocol; pp.3–5, Eqs.1–7 <i>Target:</i> External binary event <i>Output:</i> Reasoning + numeric probability	<i>Feedback / objective:</i> Observed event outcome; Log/Brier/spherical/Beta proper rewards <i>Update:</i> Group mean, no std, no KL	Forecasting; Fixed-step comparison; token budgets and seed rankings differ
Verifiable forecast rewards [70] RL / hybrid protocol; pp.4–6,12, Eq.1 <i>Target:</i> Conditional game win probability <i>Output:</i> Direct numeric span or reasoning + span	<i>Feedback / objective:</i> Historical state-bucket empirical rate; Squared error to teacher rate <i>Update:</i> GRPO with reward scaling; direct-report KL .01; masked-report KL 0 and final-span credit	Forecasting; Empirical state-bucket teacher and temporal splits determine the target. The teacher uses training-outcome state buckets with hierarchical shrinkage.
Outcome-based forecasting RL [50] RL / hybrid protocol; pp.2–4,7 <i>Target:</i> External future binary event <i>Output:</i> Reasoning + probability	<i>Feedback / objective:</i> Market/resolved future outcomes; Negative Brier plus format constraints <i>Update:</i> ReMax vs GRPO vs mean-only GRPO	Forecasting / simulated trading; Temporal checks and ensembles; not live profit evidence
Rewarding Doubt [18] RL / hybrid protocol; pp.4–6,15–16 <i>Target:</i> Fixed-answer correctness <i>Output:</i> Answer-conditioned confidence number	<i>Feedback / objective:</i> Correctness labels for fixed answers; Clipped log-score confidence reward <i>Update:</i> PPO updates confidence generation; answer held as input	Answer assessment; Discrete reports, clipping and shared weights delimit the ideal score argument. Only answer-assessment trajectories are sampled for the confidence update.

B.2. Joint reports, expert feedback and communication

Method and prediction object	Supervision and update	Use and qualifications
RLCR [19] RL / hybrid protocol; pp.4,16–18; code qualification in Appendix C <i>Target:</i> Chosen-answer correctness <i>Output:</i> Answer + uncertainty analysis + number	<i>Feedback / objective:</i> Binary verifier; $z - (c - z)^2$ <i>Update:</i> Group mean without std; no KL; BNPO	Answering/abstention; Pointwise reporting incentive; joint capability separate. Released recipe also includes format and tiny report-dependent bonuses
Multi-RLCR [51] RL / hybrid protocol; pp.3–4,19, Appendix B <i>Target:</i> Candidate-specific correctness marginals <i>Output:</i> Multiple answers + numeric confidences	<i>Feedback / objective:</i> Answer-set verifier; Sum correctness minus mean Brier <i>Update:</i> Group mean without std; sequence-level update	Answer sets; Marginals do not define joint set probability
DCPO [52] RL / hybrid protocol; p.6, Eqs.20–24; p.17 <i>Target:</i> Hybrid prompt/response correctness <i>Output:</i> Post-answer numeric confidence	<i>Feedback / objective:</i> Group mean mixed with per-rollout correctness; Correctness plus negative absolute error to a mixed group/instance target <i>Update:</i> Separate standardized advantages and token masks; no KL	Reasoning confidence; Implemented absolute error and stated proper-score theory have different incentives

Continued on the next page.

B.2. Joint reports, expert feedback and communication (continued)

Method and prediction object	Supervision and update	Use and qualifications
CoCA [34] RL / hybrid protocol; p.4, Eqs.6–11; pp.7–8 <i>Target:</i> Current-policy prompt solvability <i>Output:</i> Pre-answer numeric confidence	<i>Feedback / objective:</i> Current rollout success rate; Separate correctness and squared-error rewards <i>Update:</i> Segmented std-normalized advantages; no KL	Early confidence and prospective routing; Pre-report conditions later generation; demonstrated report cost differs from routing utility
CALIBER [35] RL / hybrid protocol; pp.5–8,18 <i>Target:</i> Pre-policy solvability and post-answer correctness <i>Output:</i> Pre and post numeric confidence	<i>Feedback / objective:</i> Group mean pre; instance outcome post; Two squared-error rewards + accuracy/format <i>Update:</i> No std/KL; warmup then joint	Reasoning confidence; ID evaluation split selects checkpoints. Warmup addresses weak variation in the pre-answer training signal.
Calibration-aware decision-token RL [31] RL / hybrid protocol; ACL pp.7–8, Eq.4; p.13 <i>Target:</i> Final candidate-token probabilities <i>Output:</i> Reasoning + decision token logits	<i>Feedback / objective:</i> Gold class and sampled-answer correctness; RL accuracy; local CE one-hot/uniform <i>Update:</i> Zero decision-token RL advantage plus local CE; shared parameters; no KL	Classification/moderation; Wrong-response uniform target is a heuristic; test-set checkpoint selection
SaySelf [53] RL / hybrid protocol; pp.3–5, Eq.2; p.6, Eq.4 <i>Target:</i> Chosen-answer correctness <i>Output:</i> Confidence + reflection	<i>Feedback / objective:</i> Cluster-frequency SFT; correctness feedback whose paper and current evaluator scales differ; $1 - 2(c - z)^2$ <i>Update:</i> PPO after SFT	Selective retrieval; Published ECE is a per-example absolute error; current evaluator confirms no bins and uses graded correctness
PPO-M / PPO-C [21] RL / hybrid protocol; pp.5–6, Eqs.2–3; pp.9,21–22 <i>Target:</i> Confidence relative to response quality <i>Output:</i> Answer plus confidence number	<i>Feedback / objective:</i> Preference pairs and learned reward-model quality; M: confidence-conditioned RM ranking; C: relative-quality reward modulation <i>Update:</i> RM adaptation then PPO, or online reward transformation; CDPO extension	RLHF confidence; Reward-model quality is a proxy; PPO-C is linear in reported confidence. PPO-C measures quality relative to a moving reward baseline.
RLMF [23] RL / hybrid protocol; pp.5,28,34–36 <i>Target:</i> Sentence confidence relative to sampling consistency <i>Output:</i> Sentence scores and linguistic uncertainty	<i>Feedback / objective:</i> Sampled-answer consistency plus correctness judge; Squared proxy alignment, factuality and format rewards <i>Update:</i> GRPO without std; selectively scales positive faithfulness advantages	Uncertainty expression; Proxy consistency and factual correctness are separate evaluation targets
Linguistic calibration [71] RL / hybrid protocol; pp.8–9,48–52 <i>Target:</i> Reader-induced answer belief <i>Output:</i> Long-form linguistic uncertainty	<i>Feedback / objective:</i> Reader probability model / QA answers; Reader log score + normalization penalty <i>Update:</i> PPO with KL and learned surrogates	Reader decisions; Depends on reader model and scored question

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B.2. Joint reports, expert feedback and communication (continued)

Method and prediction object	Supervision and update	Use and qualifications
RLVP [22] RL / hybrid protocol; pp.2–6, Eqs.1–4; p.10, Table 4 <i>Target:</i> Full class-label distribution intended to approximate $P(Y x)$ <i>Output:</i> Generated CoT plus JSON probability vector	<i>Feedback / objective:</i> Observed class and same-training-split XGBoost soft probabilities; teacher-correct filtering; Format and argmax correctness rewards plus $\exp\{-(1 - \alpha)D_y - \alpha D_t\}$. Here $D_y = \text{KL}(e_y \ p)$ and $D_t = \text{KL}(q \ p)$; e_y is the one-hot label and q the teacher. Teacher weight $\alpha : 1 \rightarrow 0$ <i>Update:</i> Paper: GRPO without reference-policy KL; 8 rollouts, temperature .7; public repository unavailable at inspection	Multiclass tabular and mixed text/tabular classification; full-vector downstream evaluation needed; Exponentiated KL is not a proper-score guarantee; teacher filtering scope unresolved
Verbalized Probability Distribution [8] RL / hybrid protocol; pp.2–4,14 <i>Target:</i> Chosen-answer confidence within reported vector <i>Output:</i> Full verbalized candidate vector	<i>Feedback / objective:</i> Correctness of the finally selected answer; separate validity penalty; Chosen-answer $z - (c - z)^2$ <i>Update:</i> GRPO with mean-only group centering	MCQ confidence; Full-vector output; selected-answer scalar supervision
Judging with Confidence [38] RL / hybrid protocol; pp.4,18–20 <i>Target:</i> Preference distribution <i>Output:</i> Autotator numeric probability	<i>Feedback / objective:</i> Persona-conditioned judge votes; Brier/log rewards for distribution matching <i>Update:</i> GRPO; exact normalization incompletely specified	LLM-as-a-judge; Main annotators are synthetic personas; soft-preference and majority-label metrics differ

B.3. Ranking, process and agent feedback

Method and prediction object	Supervision and update	Use and qualifications
CAPO [24] RL / hybrid protocol; pp.3–6, Eqs.5,16–20 <i>Target:</i> Correct-vs-incorrect response ranking <i>Output:</i> Mean response log probability	<i>Feedback / objective:</i> Math verifier; Pairwise logistic AUC surrogate <i>Update:</i> Custom advantage; no std/KL; response mask	Sample aggregation; Ranking score is not a correctness probability
CONQORD [57] RL / hybrid protocol; pp.4–5, Eqs.1–5; pp.7–8 <i>Target:</i> Relative confidence vs reward quality <i>Output:</i> Answer + numeric confidence	<i>Feedback / objective:</i> Preference-trained quality reward; Quality plus confidence-quality pair differences <i>Update:</i> PPO with KL	RAG triggering; Ranking/covariance objective is not proper probability elicitation
RLSR [25] RL / hybrid protocol; pp.4–6, Eqs.2.13–2.15; pp.7,18 <i>Target:</i> Confidence ranking for risk-coverage <i>Output:</i> Answer + confidence	<i>Feedback / objective:</i> Binary correctness; Harmonic rank-weighted error / AURC <i>Update:</i> Group-centered advantage, no std/KL	Selective prediction; Ranking can improve while ECE worsens; current released batch size differs from the paper

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B.3. Ranking, process and agent feedback (continued)

Method and prediction object	Supervision and update	Use and qualifications
RLCM [59] RL / hybrid protocol; pp.4–5,19 <i>Target:</i> Success of an answer forced from a reasoning prefix <i>Output:</i> Hidden-state probe probability	<i>Feedback / objective:</i> Monte Carlo forced-answer success by prefix; Probe BCE plus trajectory correctness and confidence-margin reward <i>Update:</i> Probe-only supervised gradient; GRPO policy update	Early exit and answer aggregation; Prefix success differs from step validity; margin rewards are not probability identification
DualStake [72] RL / hybrid protocol; pp.2–5; pp.13–14 <i>Target:</i> Evidence and final-answer confidence <i>Output:</i> Two confidence numbers in search trace	<i>Feedback / objective:</i> Answer-token F1 during training; evaluation conventions differ; Correctness plus linear confidence stakes <i>Update:</i> GRPO with warmup weights	Search agents; Clipped linear stake does not identify interior probabilities; metric targets differ
C3RL / CAS [37] RL / hybrid protocol; pp.3–4,13,15 <i>Target:</i> Correctness confidence / stopping score <i>Output:</i> Confidence 1–10	<i>Feedback / objective:</i> Verifier and base-model reference groups; Thresholded confidence and reference rewards <i>Update:</i> VeRL clipped RL; KL .001	Adaptive sample count; Threshold reward not proper; OOD hyperparameter selection
RefGRPO [73] behavior / boundary protocol; pp.5–6,15 <i>Target:</i> Binary post-feedback correctness assessment <i>Output:</i> Reflection bit	<i>Feedback / objective:</i> Execution-grounded task outcomes; Outcome + decaying reflection-match bonus <i>Update:</i> Std-normalized GRPO+; no KL	Self-verifier / selection; Binary self-assessment accuracy differs from continuous-probability calibration
CURE [55] RL / hybrid protocol; pp.4–6,13–14 <i>Target:</i> Atomic-claim correctness <i>Output:</i> Claim + numeric confidence + reasoning	<i>Feedback / objective:</i> Retrieved verification + teacher corrections; DPO calibration then masked factuality RL <i>Update:</i> SFT/format RL, calibration DPO, then claim-token factuality RL	Claim filtering; Claim precision, recall and confidence quality are different endpoints
CAR / MSCR [56] RL / hybrid protocol; p.5; pp.7–8 <i>Target:</i> Final tool-agent answer correctness <i>Output:</i> Numeric confidence after tool trace	<i>Feedback / objective:</i> EM outcome; Weighted Brier or margin-separated reward <i>Update:</i> Search-R1/GRPO; incomplete hyperparameters	Search/code agents; Unequal MSCR weights change elicited probability
RAC [60] RL / hybrid protocol; pp.3–6,16 <i>Target:</i> Relative correctness / evidence confidence <i>Output:</i> Numeric multimodal confidence	<i>Feedback / objective:</i> Correctness and corruption comparisons; Ranking hinge + clean/corrupt directional hinge <i>Update:</i> GRPO reward shaping; paired branches	Multimodal reasoning; Directional constraints do not identify absolute probabilities
Entropy-target RL [74] RL / hybrid protocol; pp.3,7–8 <i>Target:</i> Question-level error proxy <i>Output:</i> Separate reflection + numeric uncertainty	<i>Feedback / objective:</i> Sampling entropy mapped through held-out Platt fitting; Tolerance-clipped absolute error to teacher <i>Update:</i> GRPO uncertainty adapter; answer generation fixed	Amortized uncertainty; Question-level proxy does not automatically become answer-conditioned error probability

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B.3. Ranking, process and agent feedback (continued)

Method and prediction object	Supervision and update	Use and qualifications
C²GSPG [61] RL / hybrid protocol; pp.4–6, Eqs.5–10; pp.15–18 <i>Target:</i> Sequence-likelihood confidence <i>Output:</i> Geometric mean trajectory probability	<i>Feedback / objective:</i> Verifier; Sequence policy objective plus BCE on geometric-mean token likelihood <i>Update:</i> Centered reward divided by one minus old confidence; direct BCE; conflict gating	Reasoning; Sequence likelihood is a trained proxy, not inherently a correctness posterior

B.4. Actions and selective decisions

Method and prediction object	Supervision and update	Use and qualifications
Behaviorally calibrated RL [29] RL / hybrid protocol; pp.5–6, Eqs.3–4; pp.25–26 <i>Target:</i> Correctness confidence for threshold decisions <i>Output:</i> Answer probability or outcome-value readout	<i>Feedback / objective:</i> Final correctness and threshold-dependent utilities; Threshold-integrated Brier or clipped log-based score <i>Update:</i> PPO confidence/value variants; Confidence-CE uses GRPO; post-hoc acceptance thresholds	Selective answering; Threshold-weighting convention determines the score; claim probabilities require separate supervision
TruthRL [28] behavior / boundary protocol; pp.3–4,16 <i>Target:</i> Answer/abstain action <i>Output:</i> Answer or refusal	<i>Feedback / objective:</i> Task correctness and explicit abstention utility; Ternary correct/abstain/wrong reward <i>Update:</i> GRPO on discrete action behavior	Selective answering; Action probabilities are not explicit correctness reports
UCPO [66] behavior / boundary protocol; p.4, Eqs.3–5; pp.13–14 <i>Target:</i> Uncertainty action preference <i>Output:</i> Answer or explicit uncertain behavior	<i>Feedback / objective:</i> Right/wrong/uncertain outcome groups; Dynamic uncertainty advantage <i>Update:</i> Ternary decoupling and rollout filtering	Selective answering; Behavioral metrics; no numeric-probability calibration evidence
Intellectual humility [58] behavior / boundary protocol; pp.2–3,5 <i>Target:</i> Answer-vs-abstain action <i>Output:</i> Answer or IDK	<i>Feedback / objective:</i> Task verifier; Correct +1; wrong -1; tuned abstain reward <i>Update:</i> GRPO after optional abstention SFT	Abstention; Direct behavioral evidence; coverage and raw accuracy must accompany accepted-answer precision
BinaryPPO [75] behavior / boundary protocol; p.3, Eqs.3–9; pp.4–5 <i>Target:</i> Binary label-action probability <i>Output:</i> Two-action policy probabilities	<i>Feedback / objective:</i> Static binary class labels; Confidence-dependent shaping plus value, supervised and entropy terms <i>Update:</i> PPO-style contextual action learning; printed signs unresolved	Binary classification; Standard probability-quality evaluation is missing; retain as a boundary protocol

B.5. Numerical prediction

Method and prediction object	Supervision and update	Use and qualifications
DAR [27] RL / hybrid protocol; pp.3–4,14–17; code qualification in Appendix C <i>Target:</i> Numerical predictive distribution <i>Output:</i> Multiple generated numeric samples	<i>Feedback / objective:</i> Regression targets; Empirical CRPS with K-squared pair denominator and leave-one-out contribution rewards <i>Update:</i> GRPO centering/std after LOO; KL loss .001; valid-sample filtering and fallback	Regression; Finite-set score, LOO credit and normalized update require separate interpretation
REAL [26] RL / hybrid protocol; pp.4–5, Eq.4–7; p.8, Table 9; code qualification in Appendix C <i>Target:</i> Numeric predictive mean and correlation with ratings <i>Output:</i> Numerical token probabilities summarized as a scalar	<i>Feedback / objective:</i> Human ratings; Paper: squared mean error plus target log likelihood <i>Update:</i> Paper: LOO/std/advantage clamp and weighted direct derivative; current GPG launcher has direct coefficient0 and KL0	LLM-as-a-judge; Paper and public launcher differ; mean/correlation evidence is not distributional calibration. The paper’s direct-gradient weight is distinct from reference-policy KL.

B.6. Supervised, elicitation and post-hoc comparators

Method and prediction object	Supervision and update	Use and qualifications
ConfTuner [32] non-RL comparator; p.4, Eq.1; pp.23,26 <i>Target:</i> Semantic value of confidence number <i>Output:</i> Distribution over confidence-number tokens	<i>Feedback / objective:</i> Fixed-answer correctness; Exact expected squared error across numeric tokens <i>Update:</i> Exact differentiable expected loss over numeric confidence tokens; optional supervised language loss	Confidence reporting; Probability of reporting 0.8 is not correctness probability .8
Verbalized-confidence SFT [40] non-RL comparator; pp.4–6,16–18 <i>Target:</i> Fixed-answer correctness/subtask rate <i>Output:</i> Confidence words or percentages	<i>Feedback / objective:</i> Empirical subtask accuracies; Supervised confidence-token likelihood <i>Update:</i> Supervised confidence-token likelihood; fixed-answer protocol	Reporting; Subtask rate labels and decoding determine the report; RL is future work
Supervised correctness estimation [63] non-RL comparator; pp.4–7,10; Appendix C.1 <i>Target:</i> Correctness of a fixed generated answer <i>Output:</i> Probe or Yes/No-style language classifier	<i>Feedback / objective:</i> Labeled fixed answers; Binary classification with optional representation regularization <i>Update:</i> Frozen-feature probe or LoRA estimator; held-out temperature fitting	Answer assessment; Generator and estimator are separate model copies
Just Ask [41] elicitation comparator; pp.2–5,8 <i>Target:</i> Reported correctness of generated answers <i>Output:</i> Numeric/linguistic confidence prompts	<i>Feedback / objective:</i> Existing model; labels only for optional calibration; Prompted elicitation; optional NLL temperature fitting <i>Update:</i> No LLM update; prompt/decoding choice and post-hoc fitting	Answer assessment; Some methods change the selected answer; scaled results use calibration labels

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B.6. Supervised, elicitation and post-hoc comparators (continued)

Method and prediction object	Supervision and update	Use and qualifications
Confidence elicitation study [76] elicitation comparator; ICLR v2 pp.3–6 <i>Target:</i> Reported answer confidence and sampling-based uncertainty <i>Output:</i> Black-box prompts and aggregated samples	<i>Feedback / objective:</i> Existing generator; correctness labels for evaluation; Compare prompting, sampling and aggregation protocols <i>Update:</i> No policy update; inference-time confidence extraction	Answer assessment and selection; Calibration error and error ranking vary with elicitation protocol
CFT / RCFT [77] non-RL comparator; pp.5–6, Eq.6; p.7, Table 2 <i>Target:</i> Candidate-label probability <i>Output:</i> Candidate logits	<i>Feedback / objective:</i> Gold classification label; Answer-only CE (CFT); prompt/answer loss plus ECE surrogate (RCFT) <i>Update:</i> Supervised fine-tuning	Classification; Main CFT gains use no explicit calibration regularizer; ECE and accuracy trade off
CATTO [78] preference-optimization comparator; pp.4–5, Eqs.6–7; pp.24–26 <i>Target:</i> Token confidence, evaluated at final labels <i>Output:</i> Token/class probabilities	<i>Feedback / objective:</i> Preferred/rejected examples and probability-margin proxy; DPO plus confidence-linear surrogate regularization <i>Update:</i> Differentiable preference optimization	Classification and candidate selection; Fixed-target linear surrogate is not strictly proper; comparison changes can affect decoding. DPO changes token probabilities without online rollout-reward learning.
Label-smoothed SFT [62] non-RL comparator; pp.2,5–6,8,20 <i>Target:</i> Token distribution, evaluated on class confidence <i>Output:</i> Vocabulary and label-token probabilities	<i>Feedback / objective:</i> Instruction demonstrations; Cross-entropy to one-hot/uniform mixture <i>Update:</i> Supervised instruction tuning; efficient smoothed-CE kernel	Class confidence; SFT evidence does not establish persistence through later RL
Generalized Correctness Models [79] non-RL comparator; pp.3–4,22 <i>Target:</i> Correctness conditional on response and generator identity <i>Output:</i> External Yes/No predictor	<i>Feedback / objective:</i> Labeled histories from multiple generators; Correctness-token cross-entropy; optional spline calibration <i>Update:</i> Supervised LoRA; loss only on correctness tokens	Cross-model filtering; Leaves generation fixed; target-domain calibration labels and history freshness matter
Dual-Align [80] non-RL comparator; pp.3–5, Eqs.6–11; pp.11,13 <i>Target:</i> Class confidence after post-training <i>Output:</i> Temperature-rescaled final logits	<i>Feedback / objective:</i> Pre-trained-model distributions and layer signals; Distribution KL plus layer-entropy alignment <i>Update:</i> Unlabeled scalar-temperature fitting; generator fixed	Post-hoc class calibration; Reference-model fidelity is an assumption; theoretical bound requires qualification
Temperature scaling [33] non-RL comparator; pp.4,6 <i>Target:</i> Class-label confidence from fixed logits <i>Output:</i> Temperature-rescaled class probabilities	<i>Feedback / objective:</i> Held-out labeled outcomes; Negative log likelihood <i>Update:</i> Fit one positive temperature; generator fixed	Post-hoc thresholding; Preserves class argmax; distribution shift still requires evaluation
SALSA [67] non-RL comparator; p.2, Eq.1; p.10, Algorithm 1 <i>Target:</i> Structured classification labels <i>Output:</i> Single-pass structured classification readout	<i>Feedback / objective:</i> Supervised labels; Cross-entropy on renormalized class-token logits <i>Update:</i> Supervised LoRA; no autoregressive label rollout	Typed decisions; Interface comparator; source does not measure calibration

Continued on the next page.

B.6. Supervised, elicitation and post-hoc comparators (continued)

Method and prediction object	Supervision and update	Use and qualifications
Capability probes [36] non-RL comparator; pp.3–4,18–19,21,23 <i>Target:</i> Success under a specified decoding policy <i>Output:</i> Query-only probe	<i>Feedback / objective:</i> Repeated sampled-answer correctness; Supervised success-probability fitting <i>Update:</i> BCE-trained probe; answer policy fixed	Compute allocation; Targets policy-level capability, not a realized answer’s correctness

B.7. Analyses and controlled mechanism studies

These records develop incentive or estimator analyses and report controlled protocols. They are separated from the main landscape to avoid treating a theorem, a diagnostic comparison, and a deployable predictor as the same type of contribution.

Method and prediction object	Supervision and update	Use and qualifications
Reward-family analysis [20] analysis / controlled protocol; pp.3–4, Theorem 1; pp.31–34 <i>Target:</i> Joint chosen-answer correctness <i>Output:</i> Reasoning + integer confidence	<i>Feedback / objective:</i> Binary correctness; Properness + monotone optimal-answer reward <i>Update:</i> DrGRPO; no std; format/length rewards	Decision utility; Static reporting/answer incentives and finite optimization are distinct
Abstention-collapse analysis and report repair [65] analysis / controlled protocol; pp.2–7,20–21 <i>Target:</i> Success probability and answer gate <i>Output:</i> Discrete gate vs continuous report head	<i>Feedback / objective:</i> Binary outcome; Error penalties vs Brier + accuracy <i>Update:</i> Gate-policy analysis; forced answers plus direct Brier report-head learning	Abstention; Collapse law is conditional on gate/readout assumptions; repair uses a continuous head
Group-update calibration analysis [30] analysis / controlled protocol; pp.3–6; pp.10–11, Appendix A.1 <i>Target:</i> Stochastic external event probability <i>Output:</i> Numeric probability token/report	<i>Feedback / objective:</i> Observed stochastic outcome; Log proper score <i>Update:</i> Compare std-normalized GRPO vs no-std/PPO/RLOO	Forecasting; Outcome-weighting result has a specified stochastic setup; public-code paths differ

Appendix C. Implementation evidence and version boundaries

This audit distinguishes a paper’s mathematical formulation, its reported experimental settings, and an accessible public execution path. We inspected the registered PDFs and statically read official repositories discovered through paper or author-project links on 19 September 2026. We did not execute training or reproduce reported metrics. A checked commit is an immutable source reference, not proof of the configuration used for a published checkpoint. Paper-code differences below therefore identify evidence boundaries; they do not establish that the reported experiments were conducted incorrectly.

The accompanying claim ledger records individual page and code locators. This appendix retains the differences most relevant to interpreting objectives, gradients, and comparison settings. All PDF page references are physical pages.

C.0. Representative support boundaries

The main landscape compares the methods together. This table retains selected support boundaries and source locators for implementation-oriented reading.

Representative protocol	What the evaluated design supports	Inference requiring additional evidence
Temperature scaling [33] (PDF pp.4,6)	Fitting a scalar correction with fixed logits; positive temperature preserves the argmax	Improved generation or arbitrary distribution-shift reliability
ConfTuner [32] (PDF pp.4–5)	Direct supervised learning over a confidence-numeral support	Necessity of policy gradients or identification of the numeral distribution with correctness
Decision-token calibration [31] (ACL PDF pp.7–8,14–15)	A masked GRPO/CE intervention with calibration, ranking, and sensitivity results	A proper-score guarantee for the entire shared-parameter update
RLCR [19] (PDF pp.4–5,16–18)	Conditional reporting incentives in a joint answer/confidence objective	Global maximum mean accuracy within a constrained policy family
RLVP [22] (PDF pp.3–7)	Full verbalized class vectors trained with an expert-guided reward curriculum; ranking and calibration measurements	Propriety of an exponentiated score or calibration transfer inferred from AUC alone
Group-update analysis [30] (PDF pp.4–6,10–11)	Changed incentives from reward standardization in specified stochastic-outcome settings	The same failure for every score, probability object, or GRPO application
RLSR [25] (PDF pp.4–7)	Learning useful confidence ordering for selective prediction	Numerical probability semantics or certified risk at a new threshold
Preference-distribution RL [38] (PDF pp.4–7)	Fidelity to a declared soft annotator target	Human-population validity or equivalence to majority-label calibration
REAL [26] (PDF pp.4–5,8)	A mean/likelihood objective with sampled reasoning and a possible direct-gradient component	Correlation as probability calibration or equivalence of paper and released configurations
DAR [27] (PDF pp.4,17)	Empirical CRPS, leave-one-out credit, and a group-transformed update	Automatic preservation of population-score incentives through finite ensembles

C.1 RLCR: the principal recipe agrees, but the reward has additional terms

RLCR reports group centering without reward-standard-deviation division, no KL penalty, and BNPO aggregation over active tokens in a local batch [19], pp.5,18. The public [configuration](#) sets `beta=0` and `scale_rewards=False`. Its local [arguments definition](#) defaults to BNPO; the [loss branch](#) implements that aggregation.

The implemented [Brier reward](#) is $1 - (z - c)^2$, with zero for invalid formatting. The Hotpot recipe weights format, accuracy, and Brier rewards by .5 each, and assigns two confidence-related bonuses weights of 10^{-6} . For valid formatting, its main component is a positive affine transformation of correctness-plus-Brier. The total reward is nevertheless not exactly the bare formula: report-dependent bonuses and formatting rules remain. The current recipe also uses warmup .05 rather than the paper’s .1. These qualifications preserve the main mechanistic comparison while avoiding an exact-equivalence claim unsupported by the configuration.

C.2 Reward-family experiments: missing implementation evidence remains missing

Tan et al. report DrGRPO without reward standardization and with equal token weighting, together with additional format, length, and valid-confidence rewards [20], pp.31–34. Their theoretical reward family must therefore be distinguished from the complete experimental reward. For example, valid confidence contributes an additional unit reward, while length and XML-format checks contribute further bonuses.

The paper links an [anonymous code release](#), but this audit received HTTP 403 and could not inspect it. This is not evidence that code was never released. We leave actual KL, clipping, and detailed mask behavior unresolved rather than filling them from generic Unsloth defaults. Likewise, the paper’s best-effort property concerns its stated expected-reward problem, not the stability of every implementation in that reward family.

C.3 Stochastic-outcome GRPO: a controlled LM contrast and a synthetic-code qualification

The public [GRPO](#) and [no-standard-deviation](#) launchers retain the same main LM settings while switching the normalization flag [30]. Both use four rollouts, learning rate 10^{-6} , and a .001 KL loss. Their disabled KL-in-reward flag does not remove this separate KL loss.

The repository explicitly pins upstream verl. Its [corresponding advantage implementation](#) confirms the centering-versus-standardization distinction. This supports the proposed LM estimator comparison, although reproducing the reported values would require executing the complete protocol, including checkpoint selection.

A separate issue affects the released synthetic notebook. It [constructs a clipped surrogate and then overwrites the loss](#) with an unclipped ratio expression before backpropagation. Thus the current notebook does not establish that the reported synthetic clipping ablations used that clipping path. Run-specific sources would resolve the difference. The notebook’s [RLOO baseline](#) also divides the sum of other rewards by G , rather than $G - 1$. This is not the conventional leave-one-out mean, but remains independent of its own action under the appropriate sampling conditions; the naming difference alone does not establish score-gradient bias.

C.4 REAL: follow the specialized actor, not generic GRPO flags

REAL’s paper distinguishes the auxiliary likelihood coefficient λ , the direct prediction-gradient weight β , and stabilization of the trajectory term [26], pp.4–5. Its [public launcher](#) sets the direct-gradient coefficient and dedicated REAL KL coefficient to zero. The specialized actor [gates its KL calculation](#) on that dedicated coefficient. Generic GRPO KL flags elsewhere in the script therefore do not demonstrate an active REAL KL penalty.

The actual trajectory advantage also differs from the paper’s combined negative-MSE-plus-log-likelihood expression. The actor separately applies leave-one-out baselines, standardization, and clipping to [negative MSE](#) and [target-token probability](#), then combines them. Probability and log probability are not interchangeable after these transformations. The [GPG loss](#) uses $-\log \pi$ times advantage, without PPO ratio clipping. Advantage clipping is a different operation.

The REAL branch [zeros answer-token advantages](#), with direct prediction supervision controlled by the separate coefficient. This does not freeze predictions under shared parameters. The enabled remove-padding path [weights selected digit probabilities from a full-vocabulary softmax](#), without renormalizing within the digit set. Dedicated partial-group filtering also persists independently of generic filtering flags.

These findings motivate precise reporting of the paper estimator and public recipe. They do not justify claiming that the paper’s $\beta = .01$ ablation or released checkpoints were trained with the launcher’s zero default. The reported Pearson/RMSE trade-off remains author-reported evidence, rather than an independently reconstructed run.

C.5 DAR: empirical scoring, valid samples, and transformed credit

DAR’s `reward manager` agrees with the paper’s K^2 empirical-CRPS denominator and $(K - 1)^2$ leave-one-out denominator [27], p.4. This is not the $K(K - 1)$ adjustment for independent ensemble members [16].

The `GRPO implementation` subsequently centers and standardizes contribution rewards. Furthermore, `validity handling` makes the scoring ensemble size depend on successful parsing, with pointwise fallbacks for exceptional groups. Invalid outputs receive a separate worst-reward-minus-ten penalty. Thus the iid fixed- K identity is a diagnostic for the score construction, not an exact characterization of the complete training update.

The `KBSS launcher` agrees with the paper on twelve rollouts and KL loss .001, but specifies a minibatch size of 128 rather than the paper’s 16 [27], p.17. We retain that version difference rather than selecting one as the proven configuration behind the results.

C.6 ConfTuner: a directly differentiated comparator

ConfTuner’s `training computation` restricts logits to confidence tokens, normalizes them, and exactly sums their probability-weighted squared errors [32]. It backpropagates this expected loss directly. This is a supervised comparator, not policy-gradient RL over confidence reports.

When its optional supervised regularizer is enabled, the code adds that loss with coefficient one; the `example recipes` enable it for Llama and disable it for Qwen. The regularizer should not be described as explicit reference-policy KL. These examples establish an available implementation, not a run-by-run reconstruction of every reported setting.

Appendix D. Decision and System Evidence Audit

This appendix records distinctions that affect numerical comparison and reproducibility. Paper locators use physical PDF pages. Public repositories were inspected on 19 September 2026; links below fix the examined commits where available. We distinguish the published protocol, the inspected release, and unresolved correspondence between them. Static inspection establishes what code says, not that we reproduced a reported run. The observations qualify individual claims rather than rank entire papers.

D.1. Metrics with different names, targets, and implementations

SaySelf. The paper defines its “ECE” as mean individual absolute error between confidence and a binary correctness indicator, without binning [53] (p.6, Eq.4). The public `evaluation code` likewise averages `abs(confidence-correctness)/10`. Thus the absence of binning is confirmed in the inspected release. This quantity should retain an explicit definition when quoted; silently pooling it with conventional binned ECE would change the estimand.

A further distinction concerns labels. The same evaluator obtains a GPT correctness grade, reports mean grade divided by ten as accuracy, and thresholds grades at seven for AUROC. Its `default grading prompt` also conditions the grade on the presence of chain-of-thought. This differs from the paper’s binary-indicator notation. Without a historical run manifest, the release does not prove that every table used precisely that evaluator. The reported HotpotQA comparison remains interpretable with qualifications: adding RL changes the named ECE from .3704 to .3558 and reported accuracy from .3708 to .3585 [53] (p.7, Tables 1–2).

LACIE. The paper specifies nine-bin, unweighted ECE and excludes abstentions [81] (p.6). Its `evaluator` constructs `ECMetric(n_bins=9)`. However, the author-linked `metric dependency` defaults to count weighting. The inspected documentation does not pin that dependency to the paper run. The appropriate record is therefore “paper specifies unweighted ECE; current linked dependency defaults to weighted ECE,” with the historical implementation unresolved.

The population also matters. The evaluator’s `skip_none` option removes examples where either compared model gives no answer. This is a particular paired filtering rule, not an automatic property of every invocation. In the paper, Mistral-base abstention increases from .80% to 25.27%, while ECE decreases from .15 to .12 and recall from .76 to .55 [81] (p.7, Table 1). A deployment comparison should preserve the unanswered cases and their costs alongside any conditional calibration result.

DualStake. Its appendix defines ECE bin accuracy using mean answer F1, whereas the accuracy column and Brier formula use exact correctness [72] (pp.13–14). Evidence confidence and answer confidence are also elicited at different trajectory positions. These targets and prefixes should remain separate in a comparison matrix even when both scores are supervised using downstream answer quality.

The `public metric wrapper` defaults to twenty bins and passes an `is_correct` array into all three metric functions; the paper specifies ten bins. The imported `statistics/confidence_metrics.py` is absent from the inspected public tree. Accordingly, the appendix’s F1/EM distinction is source-supported, while the actual underlying metric implementation and its connection to published runs remain unresolved. The reported eight-dataset averages for DualStake are ECE .178, AUROC .712, and Brier .220; temperature scaling has lower ECE .151 but AUROC .538 and Brier .249 [72] (p.6, Table 1). These are qualified within-paper comparisons, not harmonized calibration measurements.

Just Ask for Calibration. “AUC” denotes area under selective accuracy versus coverage, rather than AUROC; the ECE description specifies squared bin error [41] (pp.2–3). No attributable implementation was located to resolve the squared-versus-absolute wording. Its temperature-scaled results use labels: one of five folds fits the temperature and the remaining four evaluate it. Learned linguistic confidence mappings instead use four folds for fitting and one for evaluation; the combined procedure separates mapping, temperature, and evaluation folds [41] (p.8). These are distinct supervised resources. On GPT-4 TriviaQA, label probabilities versus verbalized one-stage top-one reports give raw ECE .078/.024 but temperature-scaled Brier .077/.084 [41] (p.3, Table 2). The result does not support dominance under every metric.

D.2. What the objective and released artifact identify

RLSR’s paper states an effective batch of 1,536 completions and removal of reward-standard-deviation normalization [25] (p.7). The inspected `HotPotQA configuration` agrees on `scale_rewards: false` but describes 2,048 completions, from four devices, batch eight, and accumulation sixty-four. This difference belongs in artifact provenance; it does not invalidate the reported selective-prediction comparison. A reproducibility claim needs the configuration and dependency versions for the particular run, rather than the repository name alone.

Two richer output interfaces require an additional scored-object field. P067 emits a candidate vector, chooses its maximum, and rewards the selected answer’s scalar confidence, with a separate probability-sum penalty [8] (pp.2–3). Its full-vector Brier improvement is empirical evidence, not full-vector supervision; that score ranges from zero to two, unlike its main binary Brier metric [8] (p.14, Table 6). P075’s training event is an individual preference draw, its soft MSE target is a preference proportion, and its ECE/Brier target is a binarized majority after excluding ties [38] (pp.18–19). The main comparison also matches annotation count while giving RL ten times as many distinct inputs as dense-label SFT [38] (pp.5,20–21). Reporting these choices supports sensible method selection without attributing all differences to the named optimizer.

RLVP: a full-vector reward curriculum. The supplied WWW 2026 publication resolves the earlier full-text gap [22]. The policy generates CoT text and a JSON class vector; the displayed reward sums valid format, argmax correctness, and the exponential of a negative weighted sum of label-to-report and expert-to-report KL divergences (PDF pp.2–3, Eqs.1–4). The expert weight decays linearly from one to zero. This is a concrete source of denser comparisons between probability reports, but the endpoint component is p_Y , not a proper log score; Appendix A.7 gives the precise distinction. The reward specifies no semantic process supervision for CoT faithfulness.

The expert is XGBoost fitted to each task’s full training split, then used to predict that same split. The paper describes stratified 80/20 task splits and teacher-correct filtering (PDF pp.3–4). Section 2.3 restricts calculation of the probability component to teacher-correct samples, whereas Section 3.1.1 describes retaining such training samples; it is unresolved whether all losses lose the other examples or only this component is gated. The description does not establish test-label leakage, but it also does not establish cross-fitted teacher probabilities or remove teacher-selection bias. The nominal optimizer is GRPO without reference-policy KL, with eight rollouts, temperature

.7, effective batch 1,024, learning rate 2×10^{-6} , and three joint-training or six single-task epochs (PDF pp.6,10). The label-distribution KL in the reward is distinct from the omitted reference-policy KL.

The numerical evidence is metric-specific. Table 1 gives RLVP-Anneal versus XGBoost AUC wins on all six single tasks, but lower ECE on only Customer Churn (.0726 versus .1681) and Uber Analysis (.0153 versus .0431). Equal-task averages calculated from that table are AUC .9294 versus .8124 and ECE .1005 versus .0743 (PDF p.5). The other four ECE comparisons favor XGBoost. Hence the adjacent prose’s best-average-ECE description is not supported by these rows. The paper defines macro one-vs-rest AUC, summed multiclass Brier, and a ten-bin weighted absolute confidence/accuracy ECE (PDF pp.5–6, Eqs.5–7); the ECE formula is not a full-vector calibration test.

Joint training uses Qwen2.5-7B-Instruct and Llama-3.1-8B-Instruct on 169 UniPredict training splits. The reported up-to-55% Qwen and 50% Llama teacher-surpass ratios concern AUC on those tasks’ test splits; Figure 3’s mean gain is conditional on the won tasks (PDF pp.6–7). On eight unseen AMLB tasks, Table 2 compares RLVP with zero or 32 in-context examples to XGBoost trained/tuned on the same 32 examples. The win counts are four/eight and five/eight; adding examples improves RLVP AUC on three tasks and decreases it on five. That table contains no held-out ECE or Brier, so the claim of calibration transfer to unstructured text remains unsupported by that table. The protocol does not fully specify demonstration exclusion from evaluation, tuning partitions, seed dispersion, or all text preprocessing.

The adapted RLCR baseline’s displayed plus-Brier expression leaves reward orientation and scalar/vector notation unresolved (PDF p.5); this should not be interpreted as proof of an implementation error. Likewise, format rules name tags, JSON parsing, and a 10^{-5} sum tolerance without fully specifying invalid-output fallback or all component constraints. The [paper-linked repository](#) and its public GitHub API endpoint returned HTTP 404 at inspection on 19 September 2026. This means no attributable code was available to this audit, not that a repository can never become available. The source paper and our arithmetic, rather than a verified implementation or reproduced run, support the comparisons above. Appendix A.3 reports approximately 48 hours on 32 A100-80GB GPUs for a three-epoch joint run; this resource scope should accompany comparisons with task-specific tabular models.

D.3. Eight industrial sources and their evidential roles

The following table cites the original pages or repositories, not local reading notes. Entries describe the inspected source as of the access date. These artifacts are useful engineering evidence, with different disclosure levels and statistical targets.

Source	Disclosed approach or interface	What the evidence supports
Mantic technical post [101]	Brier-reward forecasting with prepared research context, reasoning, and distribution tools; centered advantages without standard-deviation division	A documented industrial training case with external event outcomes; information gathering and training contribute jointly
Jev introduction [1]	Parallel typed decisions and probability outputs; workflow evaluation against strong-model reference probabilities	Product interface and reported system comparisons; reference-model agreement is not independently labeled correctness
TypeSafe primer [90]	RLCD expanded as Reinforcement Learning for Calibrated Decisions; calibration described through group frequencies	Official terminology and intended statistical contract; no complete reproducible training algorithm
TypeSafe confidence documentation [91]	Option/level probabilities plus a distribution-shape confidence statistic	Distinct API fields whose semantics must be checked before choosing a threshold

Source	Disclosed approach or interface	What the evidence supports
harshatheg/Qwen-2.5-1B-RLCD [92]	Shared prefill, candidate-logit readout and programmatic assembly using Qwen2.5-1.5B-Instruct-4bit	An inference-engine proposal; normalized candidate scores and the artifact name do not establish calibration-aware RL training
rorshopping/jev-on-a-laptop [93]	Unofficial stock-model experiments with parallel finite-field decisions	A local interface comparator; its own documentation distinguishes raw candidate softmax from calibrated probabilities
kikoncuo/jevfire [94]	Existing LM-head scoring of verified single-token labels, shared-prefix reuse, and JSON assembly without retraining	Serving and schema evidence; independent field scoring differs from general sequential structured generation
AlexWortega/openjev [95]	Last-token sequence classification trained by cross-entropy over entailment, contradiction, and neutral	A supervised trained comparator, distinct from inference wrappers and the undisclosed official RL recipe

The community names therefore denote at least two different interventions: changing inference around existing weights and training a classifier. Neither establishes identity with Jev’s proprietary method. Similarly, structural validity prevents unsupported output forms but does not ensure that an allowed answer is correct. The useful comparison separates model training, probability readout, and output serialization.

D.4. Controls for an industrial decision benchmark

A benchmark should first fix the semantic task: candidate sets, per-field labels, abstention behavior, and whether fields are independent or condition on preceding decisions. Otherwise a parallel field scorer and a sequential generator may solve different problems. Candidate verbalizers, shared token prefixes, fallback continuation, and invalid-output handling affect both computation and the resulting scores. Preserve the same input evidence and reasoning allowance where the comparison permits; state the differences where it does not.

Serving measurements should disclose weights, precision, hardware, engine, batching, concurrency, context length, and cache state. Separate prefill, scoring, assembly, and end-to-end latency; distinguish cold model loading from fresh-prefix requests on warm kernels. Match constrained-output baselines rather than comparing finite-choice scoring only with unrestricted prose. JEVfire’s source itself separates cache conditions and notes that engine scheduling can require multiple batches; its game demonstration includes an explicit physics guard [94]. Controller rules, retrieval, and verification should therefore be ablated or held fixed when attributing performance to the model.

Statistical evaluation should use independent task outcomes, or explicitly identify a reference-model target. Report accuracy, proper loss, stated calibration diagnostics, coverage, and utility under declared costs; evaluate both cost-derived and validation-selected thresholds. Multiple decisions from one workflow are not automatically independent test cases. For subjective labels, preserve the annotator population and distinguish individual preferences from majority decisions. For forecasting, record information cutoffs, event grouping, and access to retrieval. Mantic reports a Metaculus baseline-score increase from 38.6 to 45.8, where the score rescales log probability rather than measuring ECE [101] (Results). That claim concerns the described forecasting system, not an isolated causal effect of the reward formula.

These controls make positive engineering results easier to interpret: reduced serialization work, better classification after supervised training, improved numerical probabilities, and better decisions can each be valuable. A comparison should establish which changed, what resources it required, and whether the gain survives the application’s chosen costs and information constraints.

Appendix E. Neighboring surveys and protocol qualifications

This appendix documents the scope comparisons and selected protocol details behind Sections 2 and 6. The comparison concerns the question each source organizes, not a ranking of survey quality. Page locators refer to physical PDF pages starting at 1; publication and arXiv versions are distinguished where relevant.

E.1 Scope of neighboring surveys

Survey and inspected version	Established coverage relevant to this article	More specific question pursued here
Geng et al., <i>A Survey of Confidence Estimation and Calibration in Large Language Models</i> [9], arXiv v2	Relative versus absolute confidence; generation and classification; estimation, calibration, selective generation, and human variation (PDF pp.2–9)	How a declared probability target interacts with a joint answer objective and implemented policy update
Pan et al., <i>Reward Modeling for Reinforcement Learning-Based LLM Reasoning</i> [10], arXiv v2	Reward architecture, granularity and semantics; PPO/DPO/GRPO/DAPO; uncertainty penalties, reward hacking, and downstream evaluation (PDF pp.7–19,37–43)	Whether a reporting incentive survives the transformations applied to its reward and gradient
Zhang et al., <i>From Passive Metric to Active Signal</i> [11], ACL Findings 2026	Uncertainty-guided reasoning, agents and RL; local discrimination, intrinsic rewards, process supervision, and control-oriented evaluation (PDF pp.3–9)	Connecting those uses to explicit probability targets, objective properties and estimator assumptions
Shorinwa et al., <i>A Survey on Uncertainty Quantification of Large Language Models</i> [12], arXiv v2	Token, verbal, semantic and mechanistic approaches; RL-based confidence learning, proper scores, and uncertainty-guided assistance (PDF pp.3–4,12–14,19–24)	Comparing particular scoring objectives and actual updates, rather than grouping methods by uncertainty-estimation channel
Wen et al., <i>Know Your Limits</i> [13], arXiv v3	Query answerability, model knowledge and human values; supervised/preference alignment, inference-time abstention, and coverage–quality trade-offs (PDF pp.2–11)	Distinguishing learned action preferences at specified costs from reusable probability reports
Li et al., <i>Confidence as Control</i> [14], author public copy	Decision states, units, signals, transformations and policies; training, inference, cascading, retrieval, risk and agentic control; semantics, composition and evaluation (PDF pp.2–6,22–24)	Method-level conditions connecting probability-learning incentives to implemented updates and downstream losses
Unpublished companion manuscript, <i>Beyond Natural-Language Interfaces: Decisions, Scores, and Predictive Distributions</i> , version V6	Predictive-report semantics; supervision and readout paths; classification, scoring, numerical properties and distributions; transformations, consumer losses, access and execution costs	Probability-learning incentives under joint answer objectives, practical reward and gradient transformations, and finite-ensemble scoring, connected to downstream losses

Survey and inspected version	Established coverage relevant to this article	More specific question pursued here
Li et al., <i>A Survey on the Honesty of Large Language Models</i> [102], arXiv v1; TMLR publication metadata verified	Self-knowledge and self-expression; calibration, selective prediction, and SFT/RL methods for abstention and confidence (PDF pp.3,6–7,10–14)	Comparing the statistical target, reward incentive, update, and deployment loss within confidence learning
Xia et al., <i>A Survey of Uncertainty Estimation Methods</i> [103], arXiv v2; Findings ACL 2025 metadata	Four inference-time estimation families; calibration training is explicitly outside its scope (PDF pp.2–3,6–8)	Studying interventions that train or transform probability reports and the updates that implement them
Liu et al., <i>Uncertainty Quantification and Confidence Calibration</i> [104], KDD 2025 copy	Input, reasoning, parameter and prediction uncertainty; computational cost; explicit ranking/calibration distinction (PDF pp.2–5)	Comparing reward geometry, policy updates and finite-sample objectives at the method level
Xie et al., <i>A Survey of Calibration Process for Black-Box LLMs</i> [105], arXiv v1	Estimation-to-calibration pipeline; postprocessing, auxiliary models and API access (PDF pp.2,6–8)	Connecting external calibration alternatives to the reasons for changing a policy through RL
Kang et al., <i>Uncertainty Quantification for Hallucination Detection</i> [106], arXiv v1	Signal, sampling, access and training-dependence axes for uncertainty-based detection (PDF pp.6–8,14–15)	Separating detection ranking from probability learning and decision requirements under policy changes

The comparison identifies substantial established coverage. Ranking versus calibration, confidence-guided control, supervised estimators, and RL-based abstention already appear across these surveys. We retain the primary statistical definitions in Section 2 when terminology differs between surveys; for example, AUROC measures discrimination and a min–max transformation alone does not establish probability reliability. In particular, *Confidence as Control* includes training-data selection, preference and policy optimization, reward-model uncertainty, and calibrated abstention (PDF pp.4–6, Table 1). Its general control loop is closely related to the five-component reading aid used here. The distinction of the present synthesis rests on the assumptions and evidence connecting stages, not on that diagram or the presence of an RL section.

Both *Confidence as Control* and the unpublished companion manuscript have overlapping authorship with this article. The latter was inspected as an unpublished manuscript and is included here for scope disclosure, outside the selected public literature collection. The shared material includes probability objects and conditioning in Section 2, proper-score and supervision comparisons in Sections 3–4, and control and evaluation principles in Sections 5–7. The companion organizes reports, production paths and consumer requirements; this article follows particular learning incentives through implemented updates to decision losses. Neither companion excludes training, and the distinction does not require the three surveys to cover disjoint literatures.

The resulting analysis combines existing theory and methods: proper-score reporting incentives, constrained joint-policy objectives, practical gradient transformations, and cost-specific decisions. Its contribution is the comparison of their premises and consequences, including criteria for distinguishing reporting changes from policy changes. The conditional-expectation and scoring identities used for this purpose are standard-derived synthesis rather than new general statistical theorems.

P081 is a 39-page author-hosted copy, listed as an April 2026 preprint on the author’s homepage. Acceptance and a precise arXiv identifier were not verified. The substantive reading covered PDF pp.1–6,8,11,15–19,22–25,38–39; other main-body pages were structurally scanned. Its inclusion does not imply that every reference or every version was audited. Similarly, the matrix reports the inspected versions’ coverage rather than claiming a complete census of neighboring surveys.

E.2 Publication-version checks

Decision-token calibration [31]. The original local artifact is arXiv:2601.13284v1, 16 pages. The official ACL Findings 2026 copy has 17 pages, printed pp.12537–12553. The method remains a zero decision-token advantage combined with a cross-entropy target that is one-hot for correct generations and uniform over candidates for incorrect generations (physical PDF pp.7–8, Eq.4, both versions). This hybrid update does not inherit a full-trajectory proper-score theorem merely from its calibration purpose.

The explicit test-checkpoint selection also remains: physical p.13, Appendix B in arXiv v1 and Appendix A in the ACL copy. The evaluated effect of that selection was not independently quantified. In the checked Table 4 results on physical p.8, Qwen3-4B CSQA accuracy is 81.84% for GRPO and 79.16% for the proposed method, so the result should not be summarized as per-task accuracy preservation.

The publication adds three useful evaluations:

Added ACL evidence	Exact physical locator	Interpretation
Correct/incorrect ranking	p.14, Table 7	All six reported size/task comparisons improve AUROC over GRPO; comparisons with SFT are mixed
Calibration-weight sensitivity	p.15, Appendix B.3, Table 8	The tested weights expose different in-domain and transferred calibration responses
KL sensitivity	p.15, Appendix B.4, Table 9	Limited calibration gains from the tested KL coefficient on Qwen3-1.7B

For example, the new Table 7 reports OpenAI AUROC 56.746→77.888 for Qwen3-1.7B, using a 0–100 scale. This is discrimination evidence, not measured utility at every error cost. The discussion extending the diagnosis to initially well-calibrated models is not new: it moves from arXiv p.15 to ACL p.16 and still leaves experimental validation to future work. Main-text claims using Tables 7–9 refer specifically to the ACL publication.

Active-signal survey [11]. Both the inspected arXiv:2601.15690v2 and official ACL Findings 2026 copies have 20 pages; the latter occupies printed pp.41525–41544. The checked publication passages retain local discrimination versus global calibration on physical p.3, the RL comparison on pp.6–8 and Table 3 on p.7, and the control-evaluation agenda on p.9. The limitation that no large-scale comparative experiment is provided also remains on p.9. The checked text matches after removing whitespace and page-number lines; this bounded comparison is not a certification that every aspect of the two artifacts is identical.

E.3 Selected protocol distinctions

The compact main table omits several details needed when implementing comparisons.

Protocol	Detail that changes interpretation
Fixed-answer supervision [63]	Separate answering and confidence models, roughly 20,000 labeled source examples and a 2,000-example temperature holdout; joint online training is left open (PDF pp.6,10)
EpiCaR [100]	Correct trajectories train reasoning, while successes and failures train Yes/No self-evaluation; 500 reserved test examples are excluded from final reporting, but ECE/NLL fitting descriptions conflict (PDF pp.4,17,21)

Protocol	Detail that changes interpretation
Direct actions and integrated utilities [28], [29]	A correct/abstain/wrong payoff learns a specified action trade-off; integrating threshold-dependent bounded payoffs supplies a different reporting objective (PDF pp.3–4; p.6, Eq.3)
Full-vector output [8]	Selected-answer confidence supplies the principal reward, while multiclass Brier improvements are empirical outcomes (PDF pp.2–3,14, Table 6)
Capability estimates [36]	Repeated-success targets and budget calculations depend on the decoder, sample assumptions and ability to recognize successful candidates (PDF pp.3–4,21,23)

The metric implementation audit in Appendix D gives the detailed SaySelf, DualStake and LACIE comparisons. Those findings distinguish the published definition, inspected public release and unverified historical run; no cross-paper harmonization is assumed.

Appendix F. Source versions and reproducibility

The literature snapshot is 20 September 2026. The selected collection contains 98 academic records with acquired PDFs and reading notes over stated ranges, together with eight primary web sources. It combines calibration-learning methods, necessary comparators, theoretical foundations, diagnostic studies, neighboring surveys, and decision interfaces. Source identity and reading coverage are recorded separately from independent reproduction. Publication metadata identifies verified venues where available; the table below identifies the source versions actually read.

The search was targeted to the survey scope and expanded through citation tracing and review-driven primary-source checks. Screening decisions and corpus-to-section coverage are retained with the research materials. A candidate whose full text could not be obtained, Murphy (1973), remains in the screening record and is excluded from the selected/read count and substantive citations. This access limit does not imply the paper is irrelevant.

The accompanying evidence register and revision reports retain physical source-page, equation and table locators; these locators are kept outside the main narrative. Page numbering starts at the first PDF page unless a printed-page convention is stated. Source model results are author-reported. Public-code observations refer to inspected versions rather than reconstructed historical experiment runs; no LLM training experiment was independently reproduced. The analytical examples and their numerical checks are reproducible from the supplied scripts.

The corpus was assembled on 19 September 2026 and extended through targeted publication-metadata and Brier-foundation checks on 20 September. P099 supplies the read primary source for the decomposition; the unavailable Murphy original is not promoted to a read record.

F.1. Versions used for substantive citations

Source	Read version	Version qualification
[17]	2608.28482v2	-
[70]	2607.00164v1	-
[24]	2604.12632v1	-
[51]	2603.24844v1	-
[52]	2603.09117v3	-
[34]	2603.05881v2	-
[31]	2601.13284v1	Findings of ACL 2026, July 2026
[48]	2512.25070v2	-

Source	Read version	Version qualification
[32]	2508.18847v2	Formal-publication metadata verified; the listed arXiv version remains the read source
[19]	2507.16806v2	-
[18]	2503.02623v6	-
[21]	2410.09724v2	-
[40]	2205.14334v2	-
[20]	2607.04332v1	-
[23]	2606.32032v1	-
[54]	2605.15588v2	-
[59]	2604.23333v2	-
[49]	2601.06336v2	-
[29]	2512.19920v3	ICLR 2026 non-archival workshop presentation verified; the listed arXiv version remains the read source
[28]	2509.25760v2	-
[50]	2505.17989v4	-
[77]	2505.01997v3	-
[53]	2405.20974v3	-
[57]	2404.17287v3	-
[71]	2404.00474v2	-
[72]	2609.00935v1	-
[37]	2607.01612v1	-
[65]	2608.00301v1	-
[66]	2601.22648v2	-
[25]	2607.03528v1	-
[35]	2606.24281v1	-
[73]	2606.14211v1	-
[55]	2604.12046v1	-
[58]	2601.20126v1	-
[56]	2601.07264v1	-
[60]	2605.16999v1	-
[74]	2603.06317v1	-
[30]	2508.11800v1	-
[3]	2505.14489v2	Formal-publication metadata verified; the listed arXiv version remains the read source
[2]	2303.08774v6	-
[39]	2207.05221v4	-
[7]	2606.03437v1	-
[64]	2604.16830v1	-
[4]	2603.06604v1	-
[100]	2601.06786v1	-
[6]	2505.13988v1	Formal-publication metadata verified; the listed arXiv version remains the read source

Source	Read version	Version qualification
[5]	2504.06564v3	-
[83]	2511.05993v3	-
[79]	2509.24988v2	-
[78]	2601.23096v2	-
[62]	2508.00264v2	-
[63]	2406.08391v3	-
[81]	2405.21028v2	-
[41]	2305.14975v2	-
[33]	1706.04599v2	-
[98]	2505.23854v1	-
[97]	2403.08819v2	-
[47]	2311.14648v3	-
[9]	2311.08298v2	-
[10]	2602.09305v2	-
[11]	2601.15690v2	Findings of ACL 2026, July 2026
[12]	2412.05563v2	-
[13]	2407.18418v3	-
[82]	2505.19590v5	Formal-publication metadata verified; the listed arXiv version remains the read source
[67]	2510.22691v1	-
[22]	WWW 2026 published version; pp.7401-7411; DOI 10.1145/3774904.3792540	-
[8]	2511.14275v2	-
[36]	2602.13540v1	-
[75]	2602.02708v1	-
[80]	2601.04277v1	-
[61]	2509.23129v2	-
[68]	2410.10414v2	-
[86]	2410.06707v1	-
[69]	2503.03064v2	Formal-publication metadata verified; the listed arXiv version remains the read source
[38]	2510.00263v1	-
[89]	2508.07662v1	-
[87]	2510.21049v1	-
[27]	2605.20740v1	-
[26]	2603.17145v2	-
[15]	Published JASA article, 102(477), 359–378 (2007)	-
[14]	Author-hosted 39-page preprint, retrieved 2026-09-19; exact OpenReview revision relation unverified	Public author copy; publication status unverified; recorded sections read

Source	Read version	Version qualification
[16]	Author-hosted QJRMS article, 2013 online imprint; volume 140(683), 1917–1923 (2014)	2013 online imprint; journal issue published 2014
[88]	2509.04664v1	-
[96]	2405.21047v4	Revised arXiv v4; differs from the 2024 proceedings version
[99]	2509.21882v3	-
[102]	2409.18786v1	-
[84]	2503.14476v2	Formal-publication metadata verified; the listed arXiv version remains the read source
[85]	2503.20783v2	-
[76]	2306.13063v2	-
[44]	NIPS2017 proceedings version	-
[45]	1805.08206v4	-
[46]	2107.07511v6	-
[43]	JRSSB 69(2):243–268 publisher version	-
[103]	2503.00172v2	-
[104]	2503.15850v2	-
[105]	2412.12767v1	-
[106]	2510.12040v1	-
[42]	0806.0813v2	Author arXiv v2; page anchors differ from 2009 journal pagination

F.2. Reproducibility and version boundaries

The source manifest records PDF hashes and actual reading ranges, while per-paper notes document evidence and limitations. The method matrix separates target, interface, feedback, objective, update, use, and qualification. Original manuscripts and figures remain preserved alongside the revised sources. The bibliography uses stable identifiers internally and conventional numeric references in publication exports.

The P066 paper-linked public repository was unavailable at the inspected time; its supplied publication PDF supports the method and table comparisons. P081 received recorded main-section reading and structural scanning elsewhere. These reading limits are explicit rather than treated as equal page-by-page coverage. The classical mathematical constructions demonstrate possible mechanisms; they do not measure the prevalence of those mechanisms in deployed LLMs.

F.3. Web-source persistence

Web sources are listed separately with access dates in Appendix D. Existing Wayback snapshots were queried during revision, but the availability endpoint returned HTTP 429, so no unverified archive URLs are supplied. The failed lookups are recorded, and existing repository observations retain fixed-commit links where available. Mutable documentation and a frozen code commit provide different forms of reproducibility evidence.